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Stray field immune angle sensor

Abstract

A sensor, comprising: a substrate having a reference crystal orientation and a plurality of vertical Hall element pairs that are formed on the substrate. Each vertical Hall element pair includes: (i) a respective first vertical Hall element that is oriented at a respective first angle relative to the reference crystal orientation of the substrate and (ii) a respective second vertical Hall element that is oriented at a respective second angle relative to the reference crystal orientation of the substrate. The substrate has a rectangular shape, and each of the vertical Hall element pairs is disposed in a different respective corner of the substrate.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6288533	12/2000	Haeberli et al.	N/A	N/A
9488461	12/2015	Honda et al.	N/A	N/A
9797746	12/2016	Vuillermet et al.	N/A	N/A
10782152	12/2019	Vuillermet et al.	N/A	N/A
10866117	12/2019	Kozomora et al.	N/A	N/A
10917092	12/2020	Romero	N/A	N/A
11163019	12/2020	Chaware et al.	N/A	N/A
11262422	12/2021	Romero et al.	N/A	N/A
11385075	12/2021	Foletto et al.	N/A	N/A
11555868	12/2022	Romero et al.	N/A	N/A
11561112	12/2022	Briano et al.	N/A	N/A
11733024	12/2022	Cesaretti et al.	N/A	N/A
2006/0028204	12/2005	Oohira	324/207.25	G01D 5/145
2015/0022186	12/2014	Ausserlechner	N/A	N/A
2015/0142376	12/2014	Ausserlechner	N/A	N/A
2017/0052038	12/2016	Aichriedler	N/A	G01D 5/145
2017/0248445	12/2016	Ausserlechner	N/A	N/A
2018/0340986	12/2017	Latham et al.	N/A	N/A
2020/0088546	12/2019	Sirohiwala et al.	N/A	N/A
2021/0011097	12/2020	David et al.	N/A	N/A
2021/0055129	12/2020	Kozomora et al.	N/A	N/A
2021/0072016	12/2020	Kollmitzer et al.	N/A	N/A
2021/0156664	12/2020	Binder et al.	N/A	N/A
2021/0262777	12/2020	Lassalle-Balier et al.	N/A	N/A
2022/0075008	12/2021	Romero	N/A	N/A
2022/0357144	12/2021	Tombez	N/A	N/A
2023/0003503	12/2022	Cesaretti et al.	N/A	N/A
2023/0025177	12/2022	Cesaretti et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10320057	12/2003	DE	N/A

102019124371	12/2020	DE	N/A
102020214615	12/2020	DE	N/A
3550269	12/2018	EP	N/A

OTHER PUBLICATIONS

European Intention to Grant (with Allowed Specification) dated May 2, 2024 for European Application No. EP22168908.6; 37 Pages. cited by applicant

Response to European Examination Report dated Oct. 4, 2023 for European Application No. EP22168908.6; Response filed Jan. 31, 2024; 10 Pages. cited by applicant

Communication Pursuant to Article 94(3) EPC (with pending claims) dated Oct. 4, 2023 for European Application No. 22 168 908.6-1001; 9 Pages. cited by applicant

European Response to Search or Examination Procedure for European Application No. EP22168908.6; filed Apr. 21, 2023; 17 Pages. cited by applicant

Office Action dated Nov. 29, 2021 for U.S. Appl. No. 17/015,132; 17 pages. cited by applicant

Response to Office Action dated Nov. 29, 2021, filed Feb. 22, 2022 for U.S. Appl. No. 17/015,132; 14 pages. cited by applicant

Final Office Action dated May 2, 2022 for U.S. Appl. No. 17/015,132; 15 pages. cited by applicant

Response to Final Office Action dated May 2, 2022, filed Aug. 2, 2022 for U.S. Appl. No. 17/015,132; 15 pages. cited by applicant

Advisory Action dated Aug. 23, 2022 for U.S. Appl. No. 17/015,132; 5 pages. cited by applicant

Response to Final Office Action dated Nov. 29, 2021, filed Sep. 1, 2022 for U.S. Appl. No. 17/015,132; 14 pages. cited by applicant

Advisory Action dated Sep. 22, 2022 for U.S. Appl. No. 17/015,132; 3 pages. cited by applicant

Extended European Search Report dated Oct. 12, 2022 for European Application No. EP22168908.6; 7 pages. cited by applicant

Office Action dated Dec. 29, 2022 for U.S. Appl. No. 17/363,732; 9 pages. cited by applicant

Notice of Allowance dated May 3, 2023 for U.S. Appl. No. 17/363,732; 12 Pages. cited by applicant

Response to Office Action dated Dec. 29, 2023, filed Mar. 29, 2023 for U.S. Appl. No. 17/363,732; 7 Pages. cited by applicant

European Notice of Intention to Grant dated Sep. 6, 2024 for European Application No. EP22168908.6; 35 Pages. cited by applicant

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Background/Summary

CLAIM OF PRIORITY (1) The present application is filed as a Continuation-in-Part application of U.S. application Ser. No. 17/015,132, filed on Sep. 9, 2020, and entitled Stray Field Immune Angle Sensor, which is herein incorporated by reference in its entirety. The present application is also a continuation-in-part application of U.S. application Ser. No. 17/238,543, filed on Apr. 23, 2021, and entitled Electronic Circuit Having Vertical Hall Elements Arranged on a Substrate to Reduce an Orthogonality Error, which is herein incorporated by reference in its entirety.

BACKGROUND

(1) Magnetic field sensors employ a variety of types of magnetic field sensing elements, for example, Hall effect elements and magnetoresistance elements, often coupled to a variety of

electronics, all disposed over a common substrate. A magnetic field sensing element (and a magnetic field sensor) can be characterized by a variety of performance characteristics, one of which is a sensitivity, which can be expressed in terms of an output signal amplitude versus a magnetic field to which the magnetic field sensing element is exposed. Some magnetic field sensors can detect a linear motion of a target object. Some other magnetic field sensors can detect a rotation of a target object. The accuracy with which magnetic field sensors detect an intended magnetic field can be adversely affected by the presence of stray magnetic fields (i.e., fields other than those intended to be detected).

SUMMARY

(2) According to aspects of the disclosure, a sensor is provided, comprising: a substrate having a reference crystal orientation; and a plurality of vertical Hall element pairs that are formed on the substrate, each vertical Hall element pair including: (i) a respective first vertical Hall element that is oriented at a respective first angle relative to the reference crystal orientation of the substrate and (ii) a respective second vertical Hall element that is oriented at a respective second angle relative to the reference crystal orientation of the substrate, wherein the substrate has a rectangular shape, and each of the vertical Hall element pairs is disposed in a different respective corner of the substrate.

(3) According to aspects of the disclosure, a sensor is provided, comprising: a substrate having a rectangular shape; a plurality of vertical Hall element pairs that are formed on the substrate, each of the vertical Hall element pairs being formed in a different corner of the substrate, each of the vertical Hall element pairs including: (i) a respective first vertical Hall element that is configured to face the vertical Hall element pair's respective corner, and (ii) a respective second vertical Hall element that is configured to face away from the vertical Hall element pair's respective corner.

(4) According to aspects of the disclosure, a system is provided, comprising: a ring magnet having first surface, a second surface, and a bore extending from the first surface to the second surface, the bore having a central longitudinal axis; a substrate having a reference crystal orientation; plurality of vertical Hall element pairs that are formed on the substrate, each vertical Hall element pair including: (i) a respective first vertical Hall element that is oriented at a respective first angle relative to the reference crystal orientation of the substrate and (ii) a respective second vertical Hall element that is oriented at a respective second angle relative to the reference crystal orientation of the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing features may be more fully understood from the following description of the drawings in which:

(2) FIG. 1A is a top-down view of an example of a system that includes a sensor and a ring magnet, according to aspects of the disclosure;

(3) FIG. 1B is a top-down view of the system of FIG. 1A, according to aspects of the disclosure;

(4) FIG. 1C is a side view of the system of FIG. 1A, according to aspects of the disclosure;

(5) FIG. 1D is a cross-sectional side view of the system of FIG. 1A, according to aspects of the disclosure;

(6) FIG. 2 is a schematic diagram illustrating the operation of the system of FIG. 1A, according to aspects of the disclosure;

(7) FIG. 3A is a top-down view of the sensor that is part of the system of FIG. 1A, according to aspects of the disclosure;

(8) FIG. 3B is a side view of the sensor of FIG. 3A, according to aspects of the disclosure;

(9) FIG. 3C is a side view of the sensor of FIG. 3A, according to aspects of the disclosure;

(10) FIG. 3D is a side view of the sensor of FIG. 3A, according to aspects of the disclosure;

(11) FIG. 3E is a side view of the sensor of FIG. 3A, according to aspects of the disclosure;

(12) FIG. 3F is a top-down view of the sensor of FIG. 3A, according to aspects of the disclosure;

(13) FIG. 3G is a top-down view of the sensor of FIG. 3A with the ring magnet of the system of FIG. 1A, according to aspects of the disclosure;

(14) FIG. 4A is a top-down view of a system including a ring magnet and a sensor, according to aspects of the disclosure;

(15) FIG. 4B is a top-down view of another system including a ring magnet and a sensor, according to aspects of the disclosure;

(16) FIG. 5 is a circuit diagram of the system of FIG. 4A or 4B, according to aspects of the disclosure;

(17) FIG. 6A is a top-down view of the system of FIG. 4A or 4B, according to aspects of the disclosure;

(18) FIG. 6B is a plot illustrating aspects of the operation of the system of FIG. 4A or 4B, according to aspects of the disclosure;

(19) FIG. 6C is a plot illustrating aspects of the operation of the system of FIG. 4A or 4B, according to aspects of the disclosure;

(20) FIG. 6D is a plot illustrating aspects of the operation of the system of FIG. 4A or 4B, according to aspects of the disclosure;

(21) FIG. 6E is a plot illustrating aspects of the operation of the system of FIG. 4A or 4B, according to aspects of the disclosure;

(22) FIG. 7A is top-down view of an example of a system including a magnet and a sensor, according to aspects of the disclosure;

(23) FIG. 7B is side view of an example of a system including a magnet and a sensor, according to aspects of the disclosure;

(24) FIG. 7C is cross-sectional side view of the sensor of the system of FIG. 7A, according to aspects of the disclosure;

(25) FIG. 8 is schematic diagram illustrating the operation of the system of FIG. 7A, according to aspects of the disclosure;

(26) FIG. 9A is a schematic diagram of the system of FIG. 8A, according to aspects of the disclosure;

(27) FIG. 9B is a flowchart of an example of a process, according to aspects of the disclosure;

(28) FIG. 10 is a plot of signals that are generated by using the sensor 710, according to aspects of the disclosure;

(29) FIG. 11 is a circuit diagram of a system including the sensor of FIG. 7C, according to aspects of the disclosure;

(30) FIG. 12 is a top-down view of an example of a wafer, according to aspects of the disclosure;

(31) FIG. 13A is a top-down view of an example of a sensor, according to aspects of the disclosure;

(32) FIG. 13B is a top-down view of a portion of the sensor of FIG. 13A, according to aspects of the disclosure;

(33) FIG. 13C is a top-down view of a portion of the sensor of FIG. 13A, according to aspects of the disclosure;

(34) FIG. 13D is a top-down view of a portion of the sensor of FIG. 13A, according to aspects of the disclosure;

(35) FIG. 14A is a cross-sectional side view of a system that includes the sensor of FIG. 13A, according to aspects of the disclosure;

(36) FIG. 14B is a cross-sectional top-down view of the system of FIG. 14A, according to aspects of the disclosure;

(37) FIG. 15 is a circuit diagram of the sensor of FIG. 13A, according to aspects of the disclosure;

(38) FIG. 16 is a top-down view of an example of a sensor, according to aspects of the disclosure;

and

(39) FIG. 17 is a circuit diagram of the sensor of FIG. 16, according to aspects of the disclosure.
DETAILED DESCRIPTION

(40) FIGS. 1A-D show an example of a system **100**, according to aspects of the disclosure. As illustrated, the system **100** may include a sensor **110** and a ring magnet **120**. The ring magnet **120** may include a top surface **122**, a bottom surface **124**, and a bore **126** that extends from the top surface **122** to the bottom surface **124**. The ring magnet **120** may have an inner sidewall **123** (which defines the bore **126**) and outer sidewall **125**. The ring magnet **120** may also have an inner radius **R1**, and the bore **126** of the ring magnet **120** may have a longitudinal axis B-B, as shown. The inner sidewall **123** may be symmetrical with respect to the longitudinal axis B-B (i.e., the longitudinal axis B-B can be a central longitudinal axis with the inner sidewall **123** concentric with respect to the central longitudinal axis), and the inner radius **R1** may be the distance between the inner sidewall **123** and the longitudinal axis B-B.

(41) The sensor **110** may be disposed inside the bore **126**, and subjected to a magnetic field **M** (indicated by dashed arrows in FIGS. 1A-B). The sensor **110** may include groups of Hall effect elements **112** and processing circuitry that are formed on a substrate **114**. As illustrated in FIG. 2, in operation, the ring magnet **120** may be coupled to a rotating shaft **130**, and the sensor **110** may be provided in the form of an integrated circuit (IC) mounted on a mounting member **140**. The ring magnet **120** may turn with the rotating shaft **130**, while the sensor **110** may remain in a fixed position. As a result of this arrangement, the direction of the magnetic field **M** may change (as illustrated in FIGS. 1A-B), and this change may be reflected in the signals that are generated by each Hall element group **112**. The signals that are generated by each of the Hall elements in Hall element groups **112** may be used to determine the angular position and/or speed of the rotating shaft **130** (and in some embodiments also the direction of rotation). According to the example of FIGS. 1A-E, the longitudinal axis B-B of the bore **126** is coincident with the axis of rotation of the ring magnet **120**.

(42) Hall element groups **112** may include magnetic field sensing elements that have an axis of maximum sensitivity parallel to the major, active surface (e.g., top surface **122**) of the substrate supporting the elements (e.g., Hall element groups **112** may comprise vertical Hall elements) as explained further below. Consideration of FIGS. 1A and 1B reveals that as the magnet **120** rotates, the magnetic field lines through the elements likewise rotate. The X and Y components of the magnetic field **M** are labeled **M_x** and **M_y**. Considering elements in groups **112a** and **112d** as an example, in FIG. 1A, the y-axis component of the magnetic field **M_y** is common to both groups **112a**, **112d** whereas the x-axis component of the magnetic field **M_x** is differential (i.e., equal strength on both groups **112a**, **112d** but opposite polarity). When the magnet is rotated ninety degrees as shown in FIG. 1B, the x-axis component of the magnetic field **M_x** is common to both groups **112a**, **112d** whereas the y-axis component of the magnetic field **M_y** becomes differential. Thus, combining signals from groups **112a**, **112d** will yield a differential signal that corresponds to the field **M** that is desired to be detected (i.e., the field generated by the rotating magnet). Stray magnetic fields are uniform across the sensor regardless of rotational position of the ring magnet **120** and refer generally to fields other than the rotating field **M** generated by the ring magnet. Thus, the stray magnetic field incident on groups **112a** and **112d** will always be common to both groups and thus will tend to cancel when signals from groups **112a**, **112d** are combined or subtracted. It will be appreciated that the resulting stray field immunity is cancelled or removed by placing sensing elements that form differentially processed pairs such that the field lines **M** intended to be detected are incident on the differentially processed pairs of elements in the same direction.

(43) FIGS. 3A-E show the sensor **110** in further detail. As illustrated the sensor **110** may have central axes X-X and Y-Y, which are substantially orthogonal to one another and intersect at the center **CS** of the substrate **114**. Each of the Hall element groups **112** may include a respective vertical Hall element **312** and a respective vertical Hall element **314**. Specifically, Hall element group **112a** may include a vertical Hall element **312a** and a vertical Hall element **314a**; Hall

element group **112b** may include a vertical Hall element **312b** and a vertical Hall element **314b**; Hall element group **112c** may include a vertical Hall element **312c** and a vertical Hall element **314c**; and Hall element group **112d** may include a vertical Hall element **312d** and a vertical Hall element **314d**.

(44) Vertical Hall elements are constructed from top to bottom along the depth of the substrate **114** and can be oriented to sense X, Y, or other directions parallel to a major, active surface **118** of the substrate **114** (i.e., semiconductor die) in which they are formed. Stated differently, vertical Hall elements have an axis of maximum sensitivity parallel to the major surface **118** of the substrate **114** that supports the element (in-plane fields). It will be appreciated that the side views of FIGS. 3B-E show the vertical Hall elements with an exaggerated depth for illustration purposes and that such elements generally do not extend above the major surface of the substrate. Each of the vertical Hall elements **312** may be aligned with the axis X-X, and each of the vertical Hall elements **314** may be aligned with the axis Y-Y. Accordingly, any two vertical Hall elements **312** and **314** that are part of the same Hall element group **112** may be orthogonal (i.e., arranged at a 90-degree angle) relative to one another. For example, the axes of maximum sensitivity of vertical Hall elements **312a** and **314a** may be at a 90-degree angle relative to one another; the axes of maximum sensitivity of vertical Hall elements **312b** and **314b** may be at a 90-degree angle relative to one another; the axes of maximum sensitivity of vertical Hall elements **312c** and **314c** may be at a 90-degree angle relative to one another; and the axes of maximum sensitivity of vertical Hall elements **312d** and **314d** may be at a 90-degree angle relative to one another. This arrangement allows the signals output from the vertical Hall elements in any of the Hall element groups **112** to be in quadrature with one another, which in turn allows the signals to be easily used for determining the angular position of the ring magnet **120** (and/or shaft **130**) relative to the sensor **110**.

(45) In the example of FIGS. 3A-G each group **112** includes vertical Hall elements whose axes of maximum sensitivity are arranged at an angle relative to one another (e.g., 90 degrees). However, alternative implementations are possible in which each of the groups **112** is implemented by using any other suitable type of magnetic transducers whose axes of maximum sensitivity are at an angle relative to one another and substantially parallel to the main surface of the substrate. Such magnetic field sensing elements may include vertical Hall elements, giant magnetoresistors (GMR), tunnel magnetoresistors (TMR), and or any other suitable type of magnetic transducer having an axis of maximum sensitivity that may be substantially parallel with the plane of the substrate on which the magnetic transducer is formed, etc.

(46) As used throughout the disclosure, the phrase “magnetic field sensing element is aligned with a given axis” shall be interpreted as “a magnetic field sensing element whose axis of maximum sensitivity is aligned (e.g., substantially parallel) with the given axis”. FIG. 3F shows in further detail the relative positioning of each of the vertical Hall elements **312** and **314** with respect to the substrate **114**. As illustrated in FIG. 3F, the vertical Hall elements **312** may be disposed on the periphery of the substrate **114**. Each of the vertical Hall elements **312** may be disposed at a distance DE2 from the nearest edge of the substrate **114**, and each of the vertical Hall elements **312** may be disposed at a distance DC2 from the center CS of the substrate **114**, where $DE2 < DC2$. In some implementations, the distance DE2 may be at least 90% smaller than the distance DC2.

Additionally or alternatively, in some implementations, each of the vertical Hall elements **312** may be disposed at the very edge of the substrate **114**, in which case the distance DE2 may be very close to zero. For instance, the vertical Hall elements **312** may be formed as close to the edge(s) of the substrate **114** as the manufacturing process used permits. Each of the vertical Hall elements **314** may be disposed at a distance DE1 from the nearest edge of the substrate **114**, and each of the vertical Hall elements **314** may be disposed at a distance DC1 from the center CS of the substrate **114**, where $DE1 < DC1$. In some implementations, the distance DE1 may be at least 90% smaller than the distance DC1. Additionally or alternatively, in some implementations, each of the vertical Hall elements **312** and **314** may be disposed at the very edge of the substrate **114**, in which case the

distance DC1 may be close to zero. For instance, the vertical Hall elements 314 may be formed as close to the edge(s) of the substrate 114 as the manufacturing process used permits. In some respects, disposing the vertical Hall elements 312 and 314 on the periphery of the substrate 114 is advantageous because it may result in a higher amount of magnetic flux being incident on the vertical Hall elements 312 and 314 than if the elements were closer to the center of the substrate, which in turn could increase the sensitivity of the sensor 110 with respect to the position of the ring magnet 120.

(47) According to the present example, the distance DE1 is equal to the distance DE2. However, alternative implementations are possible in which the distance DE1 is different from the distance DE2. According to the present example, the distance DC1 is equal to the distance DC2. However, alternative implementations are possible in which the distance DC1 is different from the distance DC2.

(48) Although in the example of FIG. 3F each of the vertical Hall elements 312 is spaced by the same distance from the edge of the substrate that is nearest to the vertical Hall element 312, alternative implementations are possible in which any two of the vertical Hall elements 312 are spaced by different distances from the respective edges of the substrate 114 that are closest to the two vertical Hall elements. Although in the example of FIG. 3F each of the vertical Hall elements 314 is spaced by the same distance from the edge of the substrate that is nearest to the vertical Hall element 314, alternative implementations are possible in which any two of the vertical Hall elements 314 are spaced by different distances from the respective edges of the substrate 114 that are closest to the two vertical Hall elements.

(49) FIG. 3G shows in further detail the relative positioning of each of the vertical Hall elements 312 and 314 with respect to the ring magnet 120 when the sensor 110 is installed in the bore 126 of the ring magnet 126. As illustrated, each vertical Hall element 312 may be positioned at a distance DCM1 from the longitudinal axis B-B of the bore 126 of the ring magnet 120 and each vertical Hall element 314 may be positioned at a distance DCM2 from the longitudinal axis B-B of the bore 126 of the ring magnet 120. Furthermore, each of the vertical Hall elements 312 may be positioned at a distance DS1 from the inner sidewall 123 of the ring magnet 120, and each of the vertical Hall elements 314 may be positioned at a distance DS2 from the inner sidewall 123 of the ring magnet 120.

(50) According to the present example, the distance DS1 is equal to the distance DS2. However, alternative implementations are possible in which the distance DS1 is different from the distance DS2. According to the present example, the distance DCM1 is equal to the distance DCM2. However, alternative implementations are possible in which the distance DCM1 is different from the distance DCM2. Although in the example of FIG. 3G each of the vertical Hall elements 312 is spaced by the same distance from the inner sidewall 123 of the ring magnet 120, alternative implementations are possible in which any two of the vertical Hall elements 312 are spaced by a different distance from the inner sidewall 123 of the ring magnet 120. Although in the example of FIG. 3G each of the vertical Hall elements 312 is spaced by the same distance from the longitudinal axis B-B of the bore 126 of the ring magnet 120, alternative implementations are possible in which any two of the vertical Hall elements 312 are spaced by a different distance from the longitudinal axis B-B of the bore 126 of the ring magnet 120. Although in the example of FIG. 3G each of the vertical Hall elements 314 is spaced by the same distance from the inner sidewall 123 of the ring magnet 120, alternative implementations are possible in which any two of the vertical Hall elements 314 are spaced by a different distance from the inner sidewall 123 of the ring magnet 120. Although in the example of FIG. 3G each of the vertical Hall elements 314 is spaced by the same distance from the longitudinal axis B-B of the bore 126 of the ring magnet 120, alternative implementations are possible in which any two of the vertical Hall elements 314 are spaced by a different distance from the longitudinal axis B-B of the bore 126 of the ring magnet 120.

(51) FIG. 4A shows the sensor **110** in accordance with another implementation. In this implementation, the sensor **110** includes only Hall element groups **112a** and **112d**. As illustrated, in implementations in which the sensor **110** includes only two groups of vertical Hall elements, both groups of vertical Hall elements may be disposed on the substrate **114** in an arrangement that is asymmetrical with respect to the longitudinal axis B-B of the bore **126** of the ring magnet **120**. Hall element groups **112a** and **112d** are also in an arrangement that is asymmetrical with respect to axis X-X, but symmetrical with respect to axis Y-Y, as shown. Specifically, both groups of vertical Hall elements may be disposed adjacent to the same edge **110a** of the substrate **114** (rather than being diagonally-opposed). With this configuration, rotation of the ring magnet **120** will result in incident magnetic field line variations of the same general type described above in connection with FIGS. 1A-B, thereby achieving stray field immunity. Although there can be a cost and space advantage to using only the two groups of vertical Hall elements as shown in FIG. 4A, using all four groups **112a**, **112b**, **112c**, **112d** (FIG. 1) can provide a more symmetrical configuration

(52) According to aspects of the disclosure, positioning both pairs of sensing elements on the same side of the bore **126**, as shown—in FIG. 4A, is advantageous because it allows the sensor **110** to get fully differential signals from elements **312a-312d** and **314a-314d** respectively. If these two pairs were placed in diagonally opposed directions then no differential signals could be generated in any of the two pairs, and therefore stray field cancellation would not be possible. As can be readily appreciated, both pairs sensing elements can be on one side of the substrate as shown in FIG. 4A, above or both below the X-X axis, or alternatively both sensing element pairs can be positioned on the left or right of the Y-Y axis.

(53) In some respects, having four Hall element groups **112** in the sensor **110** is advantageous because it may provide another degree of symmetry and help increase immunity to second order effects, like mechanical stresses or on-die thermal gradients. Having, two groups **112** of vertical Hall element groups **112** however is advantageous because it could help decrease the size and/or cost of manufacturing the sensor **110**, while maintaining more than adequate immunity to second order effects.

(54) The four groups are better to increase immunity to second order effects like: mechanical stresses or on-die thermal gradients: having two pairs may only partially cancel those gradients, while having four pairs provides another degree of symmetry and therefore should cancel most gradients.

(55) In the embodiment of FIG. 4A, similar to the embodiment of FIG. 3A, the vertical Hall elements are positioned at or near the periphery of the substrate to achieve a higher level of incident magnetic flux achieved by having the elements proximate to the magnet. It will be appreciated that the effect of increased magnetic flux achieved by positioning the elements near or along the periphery of the substrate (i.e., rather than nearer to the center of the substrate) can be enhanced by reducing the inner diameter of the ring magnet, both achieving the result of positioning the transducers closer to the magnet. Another way to achieve this increased magnetic flux incident on the sensing elements is by displacing the substrate with respect to the longitudinal bore of the magnet as shown in FIG. 4B.

(56) FIG. 5 is a circuit diagram of a processing circuitry **510** that is used in conjunction with the implementation of the sensor **110** which is shown in FIG. 4A or 4B. The processing circuitry **510** may include a processing path **502a** and a processing path **502d**. The processing path **502a** may be arranged to process signals that are generated by the vertical Hall elements **312a** and **312d** and the processing path **502d** may be arranged to process signals that are generated by the vertical Hall elements **314a** and **314d**.

(57) The vertical Hall element **312a** may generate a signal **501a** that is subsequently provided to a modulator **504a**. The modulator **504a** may modulate the signal **501a** based on a frequency fchop to produce a modulated signal **505a**. The vertical Hall element **312d** may generate a signal **503a** that is subsequently provided to a modulator **506a**. The modulator **506a** may modulate the signal **503a**

based on the frequency fchop to produce modulated signal **507a**. A subtractor **508a** may subtract the modulated signal **507a** from the modulated signal **505a** to produce a differential signal **509a**, which is subsequently provided to an amplifier **510a**. As can be readily appreciated from the discussion above, subtracting the two signals from one another may cancel out the effects of stray magnetic fields that are incident on the vertical Hall elements **312a** and **312d**, resulting in signal **509a** being immune to stray field effects. The amplifier **510a** may amplify the signal **509a** to produce an amplified signal **511a**, which is subsequently provided to a demodulator **512a**. The demodulator **512a** may demodulate the amplified signal **511a** based on the frequency fchop to produce a demodulated signal **513a**, which is subsequently provided to an analog-to-digital converter (ADC) **514a**. The ADC **514a** may digitize the demodulated signal **513a** to produce a digital signal **515a**, which is subsequently provided to a filter **516a**, as may be a comb filter in embodiments. The filter **516a** may filter the digital signal **515a** to produce a filtered signal **517a**, which is subsequently provided to a CORDIC module **522**.

(58) The vertical Hall element **314a** may generate a signal **501d** that is subsequently provided to a modulator **504a**. The modulator **504d** may modulate the signal **501d** based on a frequency fchop to produce a modulated signal **505a**. The vertical Hall element **314d** may generate a signal **503d** that is subsequently provided to a modulator **506a**. The modulator **506d** may modulate the signal **503d** based on the frequency fchop to produce modulated signal **507a**. A subtractor **508d** may subtract the modulated signal **507d** from the modulated signal **505d** to produce a signal **509d**, which is subsequently provided to an amplifier **510a**. As can be readily appreciated from the discussion above, subtracting the two signals from one another may cancel out the effects of stray magnetic fields that are incident on the vertical Hall elements **314a** and **314d**, resulting ins signal **509d** being immune to stray field effects. The amplifier **510d** may amplify the signal **509d** to produce an amplified signal **511a**, which is subsequently provided to a demodulator **512a**. The demodulator **512d** may demodulate the amplified signal **511d** based on the frequency fchop to produce a demodulated signal **513a**, which is subsequently provided to an analog-to-digital converter (ADC) **514a**. The ADC **514d** may digitize the demodulated signal **513d** to produce a digital signal **515a**, which is subsequently provided to a comb filter **516a**. The comb filter **516d** may filter the digital signal **515d** to produce a filtered signal **517a**, which is subsequently provided to a CORDIC module **522**.

(59) The CORDIC module **522** may include any suitable type of processing circuitry that is configured to execute a Coordinate Rotation Digital Computer (CORDIC) algorithm or otherwise compute an arctangent function (e.g., such as by using a look-up table). According to the example of FIG. 5, the CORDIC module is configured to calculate a raw position signal based on the filtered signal **517a** and the filtered signal **517d**. The raw position signal may identify the orientation of the ring magnet **120** relative to the sensor **110**, and it may be indicative of angular displacement and/or rotational speed of the ring magnet **120** (and/or the rotating shaft **130**). In some implementations, the raw position signal may be calculated in accordance with Equation 1 below:

$$(60) \ S_{\text{raw}} = \arctan\left(\frac{\text{signal}_{517a}}{\text{signal}_{517d}}\right) \quad (\text{Eq. 1})$$

where S.sub.raw is the raw position signal, signal.sub.517a is signal **517a**, and signal.sub.517d is signal **517d**.

(61) The error correction module **524** may include any suitable type of processing circuitry for adjusting the gain and/or offset of the raw position signal that is produced by the CORDIC module **522**. In operation, the error correction module **524** may receive the raw position signal from the CORDIC module **522** and generate an adjusted signal based on the received raw position signal. The adjusted signal may be generated by adjusting the gain and/or offset of the raw position signal. The gain and/or offset of the raw position signal may be adjusted, in a well-known fashion, based on a signal **533** that is generated by a temperature sensor **532**. Additionally or alternatively, the gain

and/or offset of the raw position signal may be adjusted based on a signal **535** that is generated by a trim module **534**. The trim module **534** may be a memory that is arranged to provide (to the error correction module) one or more coefficients for adjusting the gain and/or offset of the raw position signal. However, alternative implementations are possible in which the trim module **534** includes another type of device (e.g., a humidity sensor, etc.) that is used for correcting the gain and/or offset of the raw position signal. Stated succinctly, the present disclosure is not limited to any specific method for adjusting the gain and/or offset of the raw position signal.

(62) The output module **526** may include any suitable type of communications interface for outputting the adjusted signal that is produced by the error correction module **524**. The output block may format the adjusted signal into a desired output signal format and provide the formatted signal to another device (e.g., an Engine Control Unit) that is coupled to the output module **526**. The desired format may be PWM format, Single Edge Nibble Transmission (SENT) format, a Serial Peripheral Interface (SPI) format, a Local Interconnect Network (LIN) format, a CAN (Controller Area Network) format, an Inter-Integrated Circuit (I^{sup}.2C) format to name a few non-limiting examples.

(63) FIG. 5 is provided in the context of the implementation of the sensor **110** that is shown in FIG. 4B, in which the sensor **110** is provided with two pairs of sensing elements. In the example of FIG. 5, the signal S.sub.raw is generated by taking the arctan of the quotient of signals **517a** and **517d**, where signals **517a** and **517d** are generated in accordance with equations 2 and 3 below:

$$\text{signal.sub.517a} = \text{signal.sub.312a} - \text{signal.sub.312d} \quad (2)$$

$\text{signal.sub.517d} = \text{signal.sub.314a} - \text{signal.sub.314d}$ (3) where signal.sub.312a is the signal output from sensing element **312a** (also referred to as signal **501a** in FIG. 5), signal.sub.312d is the signal output from sensing element **312d** (also referred to as signal **503a** in FIG. 5), signal.sub.314a is the signal output from sensing element **314a** (also referred to as signal **501d** in FIG. 5), and signal.sub.314d is the signal output from sensing element **314d** (also referred to as signal **503d** in FIG. 5).

(64) In the implementation shown in FIG. 3G, in which the sensor **110** is provided with four pairs of sensing elements, the signals **517a** and **517d** may be generated in accordance with equations 4 and 5 below:

$$\text{signal.sub.517a} = (\text{signal.sub.312a} + \text{signal.sub.312c}) - (\text{signal.sub.312d} + \text{signal.sub.312b}) \quad (4)$$

$$\text{signal.sub.517d} = (\text{signal.sub.314a} + \text{signal.sub.314c}) - (\text{signal.sub.314d} + \text{signal.sub.314b}) \quad (5)$$

where signal.sub.312a is the signal output from sensing element **312a**, signal.sub.312b is the signal output from sensing element **312b**, signal.sub.312c is the signal output from sensing element **312c**, signal.sub.312d is the signal output from sensing element **312d**, signal.sub.314a is the signal output from sensing element **314a**, signal.sub.314b is the signal output from sensing element **314b**, signal.sub.314c is the signal output from sensing element **314c**, signal.sub.314d is the signal output from sensing element **314d**. The signals **517a** and **517d**, which are generated in accordance with equations 4 and 5, may be used to generate a signal S_{raw}, as discussed above with respect to Equation 1. As can be readily appreciated, equations 2-5 are provided for illustrative purposes only, and they do not reflect demodulation, amplification, filtering, and/or any other signal processing that might take place.

(65) FIG. 6A illustrates in further detail the impact of three parameters of the sensor **110** on the operation of the sensor **110**. The parameters include: (i) the inner radius R1 of the ring magnet, the distance DCM1 between each of the vertical Hall elements **312a** and **312d** and the longitudinal axis B-B of the bore **126** of the ring magnet **120**, (ii) the distance DCM2, between each of the vertical Hall elements **314a** and **314d** and the longitudinal axis B-B of the bore **126** of the ring magnet **120**, (iii) the distance DS1 between each of the vertical Hall elements **312a** and **312d** and the inner sidewall **123** of the ring magnet **120**, and (iv) the distance DS2 between each of the vertical Hall elements **314a** and **314d** and the inner sidewall **123** of the ring magnet **120**.

(66) FIG. 6B is a plot **610** of curves **612** and **614**. Curve **612** represents the magnetic field strength

that would be sensed by the sensor **110** along axis X-X (i.e., represents the magnetic field signal sensed by x-axis oriented element **314a** or **314d**) when R1 is set to 10 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2 mm. Curve **614** represents the magnetic field strength that would be sensed by the sensor **110** along axis Y-Y (i.e., represents the magnetic field signal sensed by y-axis oriented element **312a** or **312d**) when R1 is set to 10 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2 mm. Together, curves **612** and **614** illustrate that the measured strength of magnetic field M would vary between +70G and -70G when R1 is set to 10 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2 mm.

(67) FIG. **6C** is a plot **620** of curves **622** and **624**. Curve **622** represents the magnetic field strength that would be sensed by the sensor **110** along axis X-X (i.e., represents the magnetic field signal sensed by x-axis oriented element **314a** or **314d**) when R1 is set to 10 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2.5 mm. Curve **624** represents the magnetic field strength that would be sensed by the sensor **110** along axis Y-Y (i.e., represents the magnetic field signal sensed by y-axis oriented element **312a** or **312d**) when R1 is set to 10 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2.5 mm. Together, curves **622** and **624** illustrate that the measured strength of magnetic field M would vary between +120G and -120G when R1 is set to 10 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2.5 mm.

(68) FIG. **6D** is a plot **630** of curves **632** and **634**. Curve **632** represents the magnetic field strength that would be sensed by the sensor **110** along axis X-X (i.e., represents the magnetic field signal sensed by x-axis oriented element **314a** or **314d**) when R1 is set to 6 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2 mm. Curve **634** represents the magnetic field strength that would be sensed by the sensor **110** along axis Y-Y (i.e., represents the magnetic field signal sensed by y-axis oriented element **312a** or **312d**) when R1 is set to 6 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2 mm. Together, curves **632** and **634** illustrate that the measured strength of magnetic field M would vary between +200G and -200G when R1 is set to 6 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2 mm.

(69) FIG. **6E** is a plot **640** of curves **642** and **644**. Curve **642** represents the magnetic field strength that would be sensed by the sensor **110** along axis X-X (i.e., represents the magnetic field signal sensed by x-axis oriented element **314a** or **314d**) when R1 is set to 5 mm, DCM1 and DCM2 are both set to 1 mm, and DS1 and DS2 are both set to 2 mm. Curve **644** represents the magnetic field strength that would be sensed by the sensor **110** along axis Y-Y (i.e., represents the magnetic field signal sensed by y-axis oriented element **312a** or **312d**) when R1 is set to 5 mm, DCM1 and DCM2 are both set to 2 mm, and DS1 and DS2 are both set to 2 mm. Together, curves **642** and **644** illustrate that the measured strength of magnetic field M would vary between +160G and -160G when R1 is set to 10 mm, DCM1 and DCM2 are both set to 2 mm, and DS1 and DS2 are both set to 2 mm.

(70) In some respects, FIGS. **6B-E** illustrate that the closer the elements are symmetrically in place to the inner sidewall **123** of the ring magnet **120**, the larger the magnetic field strength that would be incident on the sensor **110**. According to the present disclosure, it has been found that the distance between the inner sidewall **123** of the ring magnet **120** and any of the vertical Hall elements **312a**, **312d**, **314a**, and **314d** of at most 2 mm is desirable in order to obtain acceptable magnetic field levels. Furthermore, FIGS. **6B-E** further illustrate that reducing the inner radius R1 of the ring magnet **120** may have a similar effect to increasing the magnetic flux density that is incident on the sensor **110**.

(71) FIGS. **7A-C** show an example of a system **700**, according to aspects of the disclosure. As illustrated, the system **700** may include a sensor **710** and a ring magnet **720**. The ring magnet **720** may include a top surface **722**, a bottom surface **724**, and a bore **726** that extends from the top

surface **722** to the bottom surface **724**. The sensor **710** may be disposed adjacent to the bottom surface **724** of the ring magnet **720** directly above the bore **726** of the ring magnet **120**. In other implementations, the sensor **710** may be disposed adjacent to the top surface of the ring magnet **720** directly below the bore **726** of the ring magnet **120**. In implementations in which the sensor **710** is off-center from the ring magnet **120**, additional circuitry may be used to compensate for harmonic distortion resulting from the off-center positioning of the sensor **710**. The sensor **710** may include planar Hall elements **712** that are formed on a substrate **714**. The substrate **714** may include an axis X-X and an axis Y-Y. The axes X-X and Y-Y are orthogonal with each other, and they may intersect at the center CS of the substrate **714**. The planar Hall elements **712a** and **712c** may be centered on the axis X-X. And the planar Hall elements **712b** and **712d** may be centered on the axis Y-Y. It will be appreciated that planar Hall elements have an axis of maximum sensitivity orthogonal to the major, active surface (e.g., top surface **722**) of the substrate.

(72) As illustrated in FIG. **8**, the magnet **720** may be coupled to rotating shaft **730** and the sensor **710** may be mounted on a mounting member **740**. The ring magnet **720** may turn with the rotating shaft **730**, while the sensor **710** may remain fixed in position. As a result of this arrangement, the direction of the magnetic field that is generated by the ring magnet **720** may change, resulting in changes in the signals that are generated by each Hall element group **712**. As is discussed further below with respect to FIGS. **9A-10**, the signals that are generated by each of the Hall element groups **712** may be used to determine the angular position and/or speed of the rotating shaft **130** (and in some embodiments also the direction of rotation). Although in the example of FIG. **7A-C** the sensor **710** is positioned below the ring magnet **120**, alternative implementations are possible in which the sensor **710** is positioned above the ring magnet **120**. Although in the example of FIGS. **7A-C** the magnet **720** is a ring magnet, alternative implementations are possible in which the magnet **720** is a disk, or puck magnet and/or any other suitable type of magnet.

(73) In some respects, arranging the planar Hall elements **712** in this manner is advantageous because it allows the calculation of the angular position (and/or speed) of the ring magnet **720** to be simplified. A simplified approach for calculating the angular position of the ring magnet **720** based on signals generated by the planar Hall elements **712** is discussed further below with respect to FIGS. **9A-B** and **11**.

(74) FIG. **9A** shows an example of a system **900A**, according to aspects of the disclosure. The system **900A** may include the rotating shaft **730**, the ring magnet **720**, and the sensor **710**, and a processing circuitry **920** that is operatively coupled to the sensor **710**. The processing circuitry may be configured to receive signal **S1**, **S2**, and **S3**, and **S4** and generate a signal **S_OUT** based on the signals **S1**, **S2**, **S3**, and **S4**, respectively. The signal **S1** may be generated by the planar Hall element **712a**; the signal **S2** may be generated by the planar Hall element **712b**; the signal **S3** may be generated by the planar Hall element **712c**; and the signal **S4** may be generated by the planar Hall element **712d**. The signal **S_OUT** may indicate the position and/or speed of rotation of the ring magnet **720** (and/or rotating shaft **730**). The manner in which the signal **S_OUT** is generated is discussed further below with respect to FIG. **9B**.

(75) FIG. **9B** is a flowchart of an example of a process **900B** for generating the signal **S_OUT**. According to the example of FIG. **9B**, the process **900B** is performed by the processing circuitry **920**. However, alternative implementations are possible in which the process **900B** is performed by another device. Stated succinctly, the present disclosure is not limited to any specific implementation of the process **900**.

(76) At step **932**, the processing circuitry **920** receives the signal **S1** from the sensor **710**. As noted above, the signal **S1** is generated by the planar Hall element **712a**.

(77) At step **934**, the processing circuitry **920** receives the signal **S2** from the sensor **710**. As noted above, the signal **S2** is generated by the planar Hall element **712b**.

(78) At step **936**, the processing circuitry **920** receives the signal **S3** from the sensor **710**. As noted above, the signal **S3** is generated by the planar Hall element **712c**.

(79) At step **938**, the processing circuitry **920** receives the signal **S4** from the sensor **710**. As noted above, the signal **S4** is generated by the planar Hall element **712d**.

(80) At step **940**, a signal **S_A** is generated based on signals **S1-S4**. In some implementations, the signal **S_A** may be generated in accordance with equation 6 below:

$$S_A = S1 + S2 - S3 - S4 \quad (\text{Eq. 6})$$

(81) At step **940**, the processing circuitry **920** generates a signal **S_B** based on signals **S1-S4**. In some implementations, the signal **S_B** may be generated in accordance with equation 7 below:

$$S_B = S1 - S2 - S3 + S4 \quad (\text{Eq. 7})$$

(82) At step **942**, the processing circuitry **920** generates a raw position signal based on the signals **S_A** and **S_B**. The raw position signal may indicate the angular position and/or speed of rotation of the ring magnet **120** (and/or rotating shaft **730**). In some implementations, the raw position signal be generated in accordance with equation 8 below:

$$(83) \quad S_{\text{raw}} = \arctan\left(\frac{S_A}{S_B}\right) \quad (\text{Eq. 8})$$

where $S_{\text{sub.raw}}$ is the raw position signal.

(84) At step **944**, the processing circuitry **920** generates a signal **S_OUT** by adjusting the gain and/or offset of the raw position signal. The gain and offset may be adjusted in a well-known fashion based on a signal that is provided by a temperature sensor and/or other data. Although in the example of FIG. **9B** gain and offset adjustment is performed on the raw position signal, alternative implementations are possible in which gain and/or offset adjustment is performed on any of the signals **S1-S4** instead. In this regard, it will be understood that the present disclosure is not limited to any specific method for performing gain and/or offset adjustment.

(85) At step **946**, the processing circuitry **920** outputs the signal **S_OUT** to another device (not shown) that is operatively coupled to the processing circuitry **920**.

(86) FIG. **10** shows a plot **1010** of the signal **S_A** and a plot **1020** of the signal **S_B**. As illustrated, in some implementations, the signal **S_A** may have a substantially sinusoidal waveform, and the signal **S_B** may have a substantially cosinusoidal waveform. According to the example of FIGS. **9A-10**, the signals **S_A** and **S_B** are in quadrature with one another.

(87) FIG. **11** is a circuit diagram of a processing circuitry **1110** that is used in conjunction with the sensor **110**. The processing circuitry **1110** may include a processing path **1102a** and a processing path **1102d**. The processing path **1102a** may be arranged to process signals that are generated by the planar Hall elements **712a** and **712c** and the processing path **1102d** may be arranged to process signals that are generated by the planar Hall elements **712b** and **712d**.

(88) The planar Hall element **712a** may generate a signal **S1** that is subsequently provided to a modulator **1104a**. The modulator **1104a** may modulate the signal **S1** based on a frequency f_{chop} to produce a modulated signal **1105a**. The planar Hall element **712c** may generate a signal **S3** that is subsequently provided to a modulator **1106a**. The modulator **1106a** may modulate the signal **S3** based on the frequency f_{chop} to produce modulated signal **1107a**. A subtractor **1108a** may subtract the modulated signal **1107a** from the modulated signal **1107a** to produce a signal **1109a**, which is subsequently provided to an amplifier **1110a**. As can be readily appreciated, subtracting the two signals from one another may cancel out the effects of stray magnetic fields that are incident on the planar Hall elements **712a** and **712c**, resulting in a signal **1109a** that is stray field immune. The amplifier **1110a** may amplify the signal **1109a** to produce an amplified signal **1111a**, which is subsequently provided to a demodulator **1112a**. The demodulator **1112a** may demodulate the amplified signal **1113a** based on the frequency f_{chop} to produce a demodulated signal **1113a**, which is subsequently provided to an analog-to-digital converter (ADC) **1114a**. The ADC **1114a** may digitize the demodulated signal **1113a** to produce a digital signal **1115a**, which is subsequently provided to a filter **1116a**, such as a comb filter. The comb filter **1116a** may filter the digital signal **1115a** to produce a filtered signal **1117a**, which is subsequently provided to a CORDIC module **1122**.

(89) The planar Hall element **712b** may generate a signal **S2** that is subsequently provided to a modulator **1104a**. The modulator **1104d** may modulate the signal **S2** based on a frequency **fchop** to produce a modulated signal **1105a**. The planar Hall element **712d** may generate a signal **S4** that is subsequently provided to a modulator **1106a**. The modulator **1106d** may modulate the signal **S4** based on the frequency **fchop** to produce modulated signal **1107d**. A subtractor **1108d** may subtract the modulated signal **1107d** from the modulated signal **1105d** to produce a signal **1109d**, which is subsequently provided to an amplifier **1110d**. As can be readily appreciated, subtracting the two signals from one another may cancel out the effects of stray magnetic fields that are incident on the planar Hall elements **712b** and **712d**, resulting in a signal **1109d** that is immune to stray fields. The amplifier **1110d** may amplify the signal **1109d** to produce an amplified signal **1111d**, which is subsequently provided to a demodulator **1112a**. The demodulator **1112d** may demodulate the amplified signal **1113d** based on the frequency **fchop** to produce a demodulated signal **1113a**, which is subsequently provided to an analog-to-digital converter (ADC) **1114a**. The ADC **1114d** may digitize the demodulated signal **1113d** to produce a digital signal **1115d**, which is subsequently provided to a comb filter **1116d**. The comb filter **1116d** may filter the digital signal **1115d** to produce a filtered signal **1117a**, which is subsequently provided to a CORDIC module **1122**. A summation element **1118a** may add the signals **1117a** and **1117d** to produce a signal **S_A**, which is subsequently provided to the CORDIC module. A summation element **1118d** may add the subtract the signal **1117d** from the signal **1117a** to produce a signal **S_B**, which is subsequently provided to the CORDIC module.

(90) The CORDIC module **1122** may include any suitable type of processing circuitry that is configured to execute a Coordinate Rotation Digital Computer (CORDIC) algorithm or otherwise compute an arctangent function (e.g., such as by using a look-up table). According to the example of FIG. 11, the CORDIC module is configured to calculate a raw position signal based on the signals **S_A** and **S_B**. The raw position signal may identify the angular position and/or speed of rotation of the ring magnet **720** relative to the sensor **710**. In some implementations, the raw position signal may be calculated in accordance with Equation 9 below:

$$(91) \ S_{\text{raw}} = \arctan\left(\frac{S_A}{S_B}\right) \quad (\text{Eq. 9})$$

where **S.sub.raw** is the raw position signal.

(92) The error correction module **1124** may include any suitable type of processing circuitry for adjusting the gain and/or offset of the raw position signal that is produced by the CORDIC module **1122**. In operation, the error correction module **1124** may receive the raw position signal from the CORDIC module **1122** and generate an adjusted signal based on the received raw position signal. The adjusted signal may be generated by adjusting the gain and/or offset of the raw position signal. The gain and/or offset of the raw position signal may be adjusted, in a well-known fashion, based on a signal **1133** that is generated by a temperature sensor **1132**. Additionally or alternatively, the gain and/or offset of the raw position signal may be adjusted based on a signal **1135** that is generated by a trim module **1134**. The trim module **1134** may be a memory that is arranged to provide (to the error correction module) one or more coefficients for adjusting the gain and/or offset of the raw position signal. However, alternative implementations are possible in which the trim module **1134** includes another type of device (e.g., a humidity sensor, etc.) that is used for correcting the gain and/or offset of the raw position signal. Stated succinctly, the present disclosure is not limited to any specific method for adjusting the gain and/or offset of the raw position signal.

(93) The output module **1126** may include any suitable type of communications interface for outputting the adjusted signal that is produced by the error correction module **1124**. The output block may format the adjusted signal into a desired output signal format and provide the formatted signal to another device (e.g., an Engine Control Unit) that is coupled to the output module **1126**. The desired format may be PWM format, Single Edge Nibble Transmission (SENT) format, a Serial Peripheral Interface (SPI) format, a Local Interconnect Network (LIN) format, a CAN

(Controller Area Network) format, an Inter-Integrated Circuit (I^{sup}.2C) format to name a few non-limiting examples.

(94) FIG. 12 is a diagram of an example of a wafer **1200**, according to aspects of the disclosure. In the example of FIG. 12, the wafer **1200** is provided with a flat **1202** denoting the crystal orientation of the wafer **1200**. Although in the example of FIG. 12 the wafer **1200** is provided with a flat to denote the crystal orientation of the wafer **1200**, alternative implementations are possible in which the wafer **1200** is provided with a notch instead.

(95) In general, when a wafer is cut from a silicon ingot, the surface of the wafer is aligned with one of the several orientations of the crystal lattice of the ingot. This orientation is known as the crystal orientation of the wafer. The crystal orientation of a wafer is usually determined by the manufacturer and marked using a flat, such as the flat **1202**, or a notch. The crystal orientation of a wafer may be defined by the Miller index, with (100) or (110) faces being the most common for silicon. The crystal orientation of a wafer is important since many of a single crystal's structural properties are highly anisotropic, which in turn causes mechanical stresses to propagate differently across the wafer in different directions.

(96) According to the example of FIG. 12, the wafer **1200** is used to manufacture magnetic field sensors that use vertical Hall elements to detect a magnetic field. Specifically, the wafer **1200** may be divided into a plurality of sensor dies **1204**, and each of the sensor dies **1204** may have one or more vertical Hall elements that are formed thereon. The configuration of any of the sensor dies **1204** is discussed further below with respect to FIGS. 13A-D.

(97) The sensor dies **1204** may be aligned with the crystal orientation of the wafer **1200**. Each of the sensor dies **1204** may have a rectangular shape. Each of the sensor dies **1204** may have two edges that are perpendicular to the crystal orientation of the wafer **1200** and two edges that are parallel to the crystal orientation of the wafer. This is consistent with the standard practice in the industry, which involves cutting (or scoring and breaking) silicon dies along and perpendicular to the crystal orientation of the wafer of which they are formed.

(98) The sensitivity of a sensor containing vertical Hall elements may be compromised by mechanical stress that is incident on the sensor. Such mechanical stress may include stress resulting from thermal expansion, stress that is imparted on the sensor die when the die is packaged, or stress that is imparted on the sensor die when a semiconductor package containing the die is soldered onto a circuit board.

(99) In one aspect, according to the present disclosure, it has been determined that the sensitivity of vertical Hall elements would change less in response to mechanical stress when the vertical Hall elements are oriented at ± 45 degrees relative to the crystal orientation of the die on which they are formed, when the crystal orientation of the die is (110), which is the most common in the industry. Specifically, according to the present disclosure, it has been determined that magnetic field sensor configurations in which vertical Hall elements are oriented at $\pm 45^\circ$ angle relative to the crystal orientation (110) of the sensor die, on which they are formed, are less susceptible to the effects of mechanical stress (on the sensitivity of the vertical Hall elements or the sensor) than sensor configurations in which the vertical Hall elements are either parallel or perpendicular to the edges of the sensor die (in embodiments in which some of the edges of the sensor die are parallel to the crystal orientation). In other words, according to the present disclosure, it has been determined that a magnetic field sensor that contains vertical Hall elements can be made more accurate by orienting the vertical Hall elements at an angle of $\pm 45^\circ$ relative to the crystal orientation of the die of the sensor.

(100) In another aspect, according to the present disclosure, it has been determined that placing different pairs of vertical Hall elements in respective corners of a substrate, and arranging one of the elements in each pair to face the pair's respective corner while arranging the other one of the elements in the pair to face away from the respective corner is also advantageous because it may result in a higher amount of magnetic flux being incident on each pair than if the vertical Hall

element pairs were not disposed at the corners of the substrate, but were closer to the center of the substrate. In other words, positioning the vertical Hall elements in the corner of the substrate, and orienting them as described, could increase the sensitivity of a magnetic field sensor with respect to the position of a ring magnet, such as the ring magnet **1420** (shown in FIG. **14A**). An example implementation of a sensor in which the vertical Hall elements are arranged in pairs and oriented as described is shown in FIG. **14B**.

(101) An example of a sensor configuration in which the vertical Hall elements are oriented at an angle relative to the crystal orientation of the sensor die is discussed further below with respect to FIGS. **13A-D**. Examples of sensor configurations in which the vertical Hall elements are either parallel or perpendicular to the edges of the sensor die are provided above with respect to FIGS. **3A-4B**. According to the present disclosure, it has been determined that the configuration discussed further below **13A-D** can have improved resistance to stress than the configurations discussed with respect to FIGS. **3A-4B** when the vertical Hall elements are oriented at an angle of $\pm 45^\circ$ relative to the crystal orientation of the die of the sensor die.

(102) Although in the example of FIG. **12**, the wafer **1200** is a silicon wafer, alternative implementations are possible in which the wafer **1200** is formed of another material, such as sapphire for example. Stated succinctly, the present disclosure is not limited to any specific implementation of the wafer **1200**.

(103) FIG. **13A** shows an example of a sensor **1300**, according to aspects of the disclosure. As illustrated, the sensor **1300** may include pairs **1312A-D** of vertical Hall elements. Pair **1312A** may include vertical Hall elements **1302A** and **1304A**; pair **1312B** may include vertical Hall elements **1302B** and **1304B**; pair **1312C** may include vertical Hall elements **1302C** and **1304C**; and pair **1312D** may include vertical Hall elements **1302D** and **1304D**. The vertical Hall elements in each of the pairs **1312A-D** may be formed on a sensor die **1204**. Although not shown, also formed on the sensor die **1204** may be signal processing circuitry, such as the processing circuitry **1510**, which is discussed further below with respect to FIG. **15**.

(104) In some implementations, each of the vertical Hall elements in each (or at least one) of the pairs **1312A-D** may be arranged at an angle relative to the crystal orientation of the sensor die **1204**. Examples of different ways to arrange a vertical Hall element at an angle relative to the crystal orientation of a sensor die are provided further below with respect to FIG. **13B**.

(105) Additionally or alternatively, in some implementations, the vertical Hall element **1304** in each of the pairs **1312A-D** may be arranged to face a respective corner of the sensor die **1204**, while the vertical Hall element **1302** in the same pair is arranged to face away from the same corner. Examples of different ways to arrange a vertical Hall element to face a corner of a sensor die are discussed further below with respect to FIG. **13C**. Examples of different ways to arrange a vertical Hall element to face away from a corner of a sensor die are discussed further below with respect to FIG. **13D**.

(106) In the example of FIG. **13A**, the vertical Hall element **1304A** is arranged to face corner **1314A** of the sensor die **1204** and the vertical Hall element **1302A** is arranged to face away from the corner **1314A** of the sensor die **1204**. The vertical Hall element **1304B** is arranged to face corner **1314B** of the sensor die **1204** and the vertical Hall element **1302B** is arranged to face away from the corner **1314B** of the sensor die **1204**. The vertical Hall element **1304C** is arranged to face corner **1314C** of the sensor die **1204** and the vertical Hall element **1302C** is arranged to face away from the corner **1314C** of the sensor die **1204**. The vertical Hall element **1304D** is arranged to face corner **1314D** of the sensor die **1204** and the vertical Hall element **1302D** is arranged to face away from the corner **1314D** of the sensor die **1204**.

(107) Each of the vertical Hall elements **1302A-D** and **1304A-D** may have a respective axis of maximum sensitivity. The axis of maximum sensitivity of each of the vertical Hall elements **1302A-D** and **1304A-D** is denoted in FIG. **13A** as a solid black arrow that is superimposed over the vertical Hall element.

(108) In some implementations, the axes of maximum sensitivity of the vertical Hall elements **1302A-D** and **1304A-D** may be aligned with the diagonals **D1** and **D2** of the sensor die **1204**. Specifically, the axis of maximum sensitivity of vertical Hall element **1302A** may be substantially perpendicular to diagonal **D1** of the sensor die **1204**. The axis of maximum sensitivity of vertical Hall element **1304A** may be substantially parallel to diagonal **D1** of the sensor die **1204**. The axis of maximum sensitivity of vertical Hall element **1302B** may be substantially perpendicular to diagonal **D2** of the sensor die **1204**. The axis of maximum sensitivity of vertical Hall element **1304B** may be substantially parallel to diagonal **D2** of the sensor die **1204**. The axis of maximum sensitivity of vertical Hall element **1302C** may be substantially perpendicular to diagonal **D1** of the sensor die **1204**. The axis of maximum sensitivity of vertical Hall element **1304C** may be substantially parallel to diagonal **D1** of the sensor die **1204**. The axis of maximum sensitivity of vertical Hall element **1302D** may be substantially perpendicular to diagonal **D2** of the sensor die **1204**. The axis of maximum sensitivity of vertical Hall element **1304D** may be substantially parallel to diagonal **D2** of the sensor die **1204**. As illustrated, in some implementations, pairs **1312A-D** may be symmetrical with respect to the center **C** of the sensor die **1204**. As used throughout the disclosure, the term “substantially” as used in the phrase “substantially perpendicular” shall mean within “10 degrees of a right angle”.

(109) Additionally or alternatively, in some implementations, one of the vertical Hall elements in each of the pairs **1312A-D** may be offset by $+X^\circ$ from the crystal orientation of the sensor die **1204**, and the other one of the vertical Hall elements in each of the pairs **1312A-D** may be offset by $-(90-X)^\circ$, where X is a positive integer, and $X < 90$. In some implementations, the value of X may be 45° for pairs **1312A** and **1312C**, and 135° for pairs **1312B** and **1312D**.

(110) More precisely, the angle between the axis of maximum sensitivity of each of the vertical Hall elements **1302A-D** and **1304A-D** may be expressed within the framework of a unit circle **U**, which is also shown in FIG. **13A**. In that framework, line **CD** extends between 0° and 180° and denotes the crystal orientation of the sensor die **1204**. The axes of maximum sensitivity of vertical Hall elements **1302A-D** and **1304A-D** may be oriented as follows: the axis of maximum sensitivity of vertical Hall element **1304A** may be oriented at 45° , the axis of maximum sensitivity of vertical Hall element **1304B** may be oriented at 315° , the axis of maximum sensitivity of vertical Hall element may be oriented at 225° , the axis of maximum sensitivity of vertical Hall element **130D** may be oriented at 135° , the axis of maximum sensitivity of vertical Hall element **1312A** may be oriented at 315° , the axis of maximum sensitivity of vertical Hall element **1302B** may be oriented at 225° , the axis of maximum sensitivity of vertical Hall element **1302C** may be oriented at 135° , and the axis of maximum sensitivity of vertical Hall element **1302D** may be oriented at 45° . Depending on how one chooses to describe the angles, it can be readily appreciated that under the latter arrangement, the smallest angle between line **CD** and the axis of maximum sensitivity of each of the vertical Hall elements **1302A-D** and **1304A-D** is either $+45^\circ$ or -45° . Although in the example of FIG. **13A** the sensor die **1204** has a rectangular shape, alternative implementations are possible in which the sensor die **1204** has a square shape. As discussed above, arranging the vertical Hall elements at $\pm 45^\circ$ relative to the crystal orientation of the sensor die **1204** is advantageous because it could enhance the sensor's resilience to mechanical stress and make the sensor more accurate. In some implementations, when the sensor die **1204** has a square shape, the axes of maximum sensitivity of the Hall elements **1302A-D** and **1304A-D** may be either substantially perpendicular or substantially parallel to the diagonals of the die.

(111) Although, in the example of FIG. **13A**, the vertical Hall elements in each of the pairs **1312A-D** are rotated (in opposite directions) by the same amount relative to the crystal orientation of the sensor die **1204**, alternative implementations are possible in which any two vertical Hall elements that are part of the same pair **1312** are rotated in opposite directions and by different amounts from the crystal orientation of the sensor die **1204**.

(112) The vertical Hall elements **1302A-D** and **1304A-D** may be disposed in the periphery of the

sensor die **1204**. In some implementations, the distance between each (or at least one) of the vertical Hall elements **1302A-D** and **1304A-D** and the nearest edge of the sensor die **1204** may be at least 90% smaller than the distance between that vertical Hall element and the center C of the sensor die **1204**. In some implementations, the distance between each (or at least one) of the vertical Hall elements **1302A-D** and **1304A-D** and the nearest edge of the sensor die **1204** may be many times smaller (e.g., 5 times, 10 times, 20 times, etc.) than the distance between that vertical Hall element and the center C of the sensor die **1204**.

(113) Additionally or alternatively, in some implementations, the vertical Hall elements **1302A-D** and **1304A-D** may be formed as close to the edges **1315A-D** of the sensor die as permitted by manufacturing tolerances. Additionally or alternatively, in some implementations, each of the vertical Hall elements **1304A-D** may be formed as close as possible to one of the corners **1314A-D** as permitted by the manufacturing tolerances. Additionally or alternatively, in some implementations, each of the vertical Hall elements **1302A-D** may be formed as close as possible to one of the corners **1314A-D** as permitted by the manufacturing tolerances. Additionally or alternatively, in some implementations, each of the pairs **1312A-D** may be formed as close as possible to a different one of the corners **1314A-D** as permitted by the manufacturing tolerances.

(114) Although, in the example of FIG. **13A**, the vertical Hall element **1302** in each of the pairs **1312A-D** is situated between the vertical Hall element **1304** which is part of the same pair and the center C of the sensor die **1204**, alternative implementations are possible in which the positions of the vertical Hall elements in the pair are switched (while their orientations remain the same). In the latter case, the vertical Hall element **1304** in each of the pairs **1312A-D** may be positioned between the vertical Hall element **1302** in that pair and the center C of the sensor die **1204**.

(115) FIG. **13B** is a partial view of the sensor **1300**, which shows the vertical Hall elements **1302D** and **1304D** in further detail. As illustrated, vertical Hall element **1302D** may have short sidewalls **1322A** and **1322B** and long sidewalls **1324A** and **1324B**. Sidewalls **1322A** and **1322B** may be substantially parallel to each other and have the same length. Sidewalls **1324A** and **1324B** may be substantially parallel to each other and have a greater length than the sidewalls **1322A-B**. The vertical Hall element **1302D** may have a longitudinal axis A-A that is substantially parallel to the sidewalls **1324A-B**. Vertical Hall element **1304D** may have long sidewalls **1334A** and **1334B**, and short sidewalls **1332A** and **1332B**. Sidewalls **1332A-B** may be substantially parallel to each other and have the same length. Sidewalls **1334A-B** may also be substantially parallel to each other and have a greater length than the sidewalls **1332A-B**. The vertical Hall element **1304D** may have a longitudinal axis B-B that is substantially parallel to sidewalls **1324A** and **1324B**. As used throughout the present disclosure, the phrase “substantially parallel” shall mean “within 10 degrees of being exactly parallel”.

(116) Further shown in FIG. **13B** is a line CD that has the same direction as the crystal orientation of the sensor die **1204**. The longitudinal axis A-A of the vertical Hall element **1302D** forms an angle **1321** with line CD (or with the crystal orientation of the sensor die **1204**). The longitudinal axis B-B of the vertical Hall element **1304D** forms an angle **1323** with line CD (or with the crystal orientation of the sensor die **1204**). As discussed above, angles **1321** and **1323** may each be equal to 45°—in other words, in the example of FIG. **13B**, the vertical Hall elements **1302D** and **1304D** are rotated by the same amount and in opposite directions from the line CD. However, it will be understood that alternative implementations are possible in which the vertical Hall elements **1302D** and **1304D** are rotated by different amounts (and in opposite directions) relative to the crystal orientation of the sensor die **1204**, in which case the values of angles **1321** and **1323** would differ. Stated succinctly, the present disclosure is not limited to any specific values for angles **1321** and **1323**.

(117) In the example of FIG. **13B**, the vertical Hall element **1302D** is orientated at an angle relative to the crystal orientation of the sensor die **1204**. In one implementation, the vertical Hall element **1302D** may be considered to be oriented at an angle relative to the crystal orientation of the sensor

die **1204** when the angle **1321** exceeds a predetermined threshold **T1** while remaining below a threshold **T2** (and/or when the angle between one of the sidewalls **1324A** and **1324B** is between thresholds **T1** and **T2**). By way of example, the threshold **T1** may be equal to one of 5°, 15°, 25°, 35°, 45°, 55°, 65°, 75°, or 85°, and the threshold **T2** may be equal to 5°. In other words, the vertical Hall element **1302D** may be orientated at an angle relative to the crystal orientation of the sensor die **1204** when the longitudinal axis A-A (and/or at least one of the sidewalls **1324A-B**) is neither parallel (e.g., substantially parallel) nor perpendicular (e.g., substantially perpendicular) to the crystal orientation of the sensor die **1204**.

(118) In the example of FIG. **13B**, the vertical Hall element **1304D** is orientated at an angle relative to the crystal orientation of the sensor die **1204**. In one implementation, the vertical Hall element **1304D** may be considered to be oriented at an angle relative to the crystal orientation of the sensor die **1204** when the angle **1323** exceeds a predetermined threshold **T3** while remaining below a threshold **T4** (and/or when the angle between one of the sidewalls **1324A** and **1324B** is between thresholds **T1** and **T2**). By way of example, the threshold **T3** may be equal to one of 35°, 45°, 55°, 65°, 75°, 85°, and the threshold **T4** may be equal to 35°. In other words, the vertical Hall element **1304D** may be orientated at an angle relative to the crystal orientation of the sensor die **1204** when the longitudinal axis B-B (and/or at least one of the sidewalls **1334A-B**) is neither parallel (e.g., substantially parallel) nor perpendicular (e.g., substantially perpendicular) to the crystal orientation of the sensor die **1204**.

(119) In some implementations, each of the vertical Hall elements **1302A-C** may have the same or similar configuration as vertical Hall element **1302D**. Additionally or alternatively, in some implementations, each of the vertical Hall elements **1304A-C** may have the same or similar configuration as the vertical Hall element **1304D**. Under ideal circumstances, the axes of maximum sensitivity of each of the vertical Hall elements **1302A-D** and **1304A-D** are perpendicular to the longitudinal axis and/or at least one of the “long” sides of the vertical Hall element. In this regard, any description of the orientation of the axis of maximum sensitivity of a vertical Hall element is also intended (in the alternative) to describe the position in space of the physical dimensions of the vertical Hall element, irrespective of whether the axis of maximum sensitivity ends up in practice being perpendicular to the long walls or longitudinal axis of the vertical Hall element. The examples provided with respect to FIGS. **12-17** assume that the two of the edges of each sensor die **1204** are parallel to the crystal orientation of the sensor die, and the other two are perpendicular. However, it will be understood that the present disclosure is not limited thereto.

(120) FIG. **13C** is a partial view of the sensor **1300**, which shows the vertical Hall element **1304D** in further detail. As illustrated, the vertical Hall element **1304D** may be positioned on the sensor die **1204** in such a way that the diagonal **D2** of the sensor die **1204** runs through the middle **M1** of the sidewall **1334A** and through the middle **M2** of the sidewall **1334B**. The diagonal **D2** may form an angle **1325** with the longitudinal axis B-B of the vertical Hall element **1304D**. In the example of FIG. **13C** the vertical Hall element **1304D** is arranged to face the corner **1314D** of the sensor die **1204**. In some implementations, the vertical Hall element **1304D** may be considered to be facing the corner **1314D** if the angle **1325** is greater than 15° (or another threshold value such as 25° or 35°). Additionally or alternatively, in some implementations, the vertical Hall element **1304D** may be considered to be facing the corner **1314D** if the diagonal **D2** runs through the middle of both (or at least one) of sidewalls **1334A-B**. Additionally or alternatively, in some implementations, the vertical Hall element **1304D** may be considered to be facing the corner **1314D** if the sidewalls **1332A** and **1332B** are disposed on opposite sides of the diagonal **D2**.

(121) FIG. **13D** is a partial view of the sensor **1300**, which shows the vertical Hall element **1302D** in further detail. As illustrated, the vertical Hall element **1302D** may be positioned on the sensor die **1204** in such a way that the diagonal **D2** of the sensor die **1204** runs through the middle **M1** of the sidewall **1322A** and through the middle **M2** of the sidewall **1322B**. In the example of FIG. **13D**, the vertical Hall element **1302D** is arranged to face away from the corner **1314D** of the sensor die

1204. In some implementations, the vertical Hall element **1302D** may be considered to be facing away from the corner **1314D** if the diagonal **D2** runs through the middle of both (or at least one) of sidewalls **1322A-B**. Additionally or alternatively, in some implementations, the vertical Hall element **1302D** may be considered to be facing away from the corner **1314D** if the sidewalls **1324A** and **1324B** are disposed on opposite sides of the diagonal **D2**. Additionally or alternatively, in some implementations, the vertical Hall element **1302D** may be considered to be facing away from the corner **1314D** if the angle between the diagonal **D2** and at least one of the sidewalls **1324A-B** is less than a threshold. The threshold may be equal to 35°, 45°, etc. Additionally or alternatively, in some implementations, the vertical Hall element **1302D** may be considered to be facing away from the corner **1314D** if the angle between the diagonal **D2** and the longitudinal axis **A-A** (shown in FIG. **13A**) is less than a threshold. The threshold may be equal to 5°, 15°, etc.

(122) FIGS. **14A-B** are diagrams of an example of a system **1400**, according to aspects of the disclosure. Specifically, FIG. **14A** shows a cross-sectional side view of the system **1400** taken along an axis **C2-C2** and FIG. **14B** shows a cross-sectional top-down view of the system **1400** taken along an axis **C1-C1**. As illustrated, the system **1400** may include a ring magnet **1420** that is coupled to a rotating shaft **1430**. The ring magnet **1420** may be configured to rotate with the rotating shaft **1430**. The sensor **1300** may be disposed on a mounting member **1440**. The sensor **1300** may be inserted in a bore **1426** of the ring magnet **1420**. As illustrated, the bore **1426** may extend between a top surface **1427** and a bottom surface **1429** of the ring magnet **1420**. In operation, the rotating shaft **1430** may rotate, and the ring magnet **1420** may rotate together with the rotating shaft **1430**. As the ring magnet **1420** rotates, the magnetic field that is produced by the ring magnet **1420** would also rotate, causing the magnetic flux incident on each of the vertical Hall elements in the sensor **1300** to oscillate. At any given point in time, each of the vertical Hall elements in the sensor **1300** may output a signal that is indicative of the magnitude of the magnetic flux incident on that vertical Hall element. As is discussed further below with respect to FIG. **15**, the signals generated by the vertical Hall elements in the sensor **1300** may be used to determine the angular position of the ring magnet. Although in the example of FIGS. **14A-B** the sensor **1300** is inserted in the bore **1426** of the ring magnet **1420**, alternative implementations are possible in which the sensor **1300** is situated above or below the magnet **1420**.

(123) FIG. **15** is a diagram of a processing circuitry **1510** that can be included in the sensor **1300** (shown in FIG. **13**). The processing circuitry **1510** may include a processing path **1502a** and a processing path **1502b**. The processing path **1502a** may be arranged to process signals that are generated by the vertical Hall elements **1304A-D** and the processing path **1502b** may be arranged to process signals that are generated by the vertical Hall elements **1302A-D**.

(124) Processing path **1502a** may include vertical Hall elements **1304A-D**, modulators **1505a-d**, subtraction elements **1511a-b**, a summation element **1513a**, an amplifier **1510a**, a demodulator **1512a**, an analog-to-digital converter (ADC) **1514a**, and a comb filter **1516a**. Vertical Hall element **1304A** may generate a signal **1501a**, which is subsequently modulated by the modulator **1505a**. Vertical Hall element **1304D** may generate a signal **1501d**, which is subsequently modulated by the modulator **1505d**. Vertical Hall element **1304B** may generate a signal **1501b**, which is subsequently modulated by the modulator **1505b**. Vertical Hall element **1304C** may generate a signal **1501c**, which is subsequently modulated with the modulator **1505c**. The demodulation of each of the signals **1501a-d** may be performed based on a chopping frequency f_{chop} . After the signals **1501a-d** are demodulated, modulated signal **1501d** may be subtracted from modulated signal **1501a** by using the subtraction element **1511a**. modulated signal **1501c** may be subtracted from modulated signal **1501b** by using the subtraction element **1511b**. The summation element **1513a** may then add the difference of signals **1501d** and **1501a** with the difference of signals **1501c** and **1501b** to produce a signal **1517a**. The amplifier **1510a** may amplify the signal **1517a**. The modulator **1512a** may modulate the amplified signal **1517a** based on the chopping frequency f_{chop} . The ADC **1514a** may digitize the modulated signal **1517a**. And the comb filter **1516a** may filter the digitized signal

1517a.

(125) Processing path **1502b** may include vertical Hall elements **1302A-D**, demodulators **1507a-d**, subtraction elements **1511c-d**, a summation element **1513b**, an amplifier **1510b**, a modulator **1512b**, an analog-to-digital converter (ADC) **1514b**, and a comb filter **1516b**. Vertical Hall element **1302A** may generate a signal **1503a**, which is subsequently modulated by the modulator **1507a**. Vertical Hall element **1302D** may generate a signal **1503d**, which is subsequently modulated by the modulator **1507d**. Vertical Hall element **1302B** may generate a signal **1503b**, which is subsequently modulated by the modulator **1507b**. Vertical Hall element **1302C** may generate a signal **1503c**, which is subsequently modulated by the modulator **1507c**. The demodulation of each of the signals **1503a-d** may be performed based on a chopping frequency f_{chop} . After the signals **1501a-d** are demodulated, modulated signal **1503d** may be subtracted from modulated signal **1503a** by using the subtraction element **1511c**. modulated signal **1501c** may be subtracted from modulated signal **1501b** by using the subtraction element **1511d**. The summation unit **1513b** may then add the difference of signals **1503d** and **1503a** with the difference of signals **1503c** and **1503b** to produce a signal **1517b**. The amplifier **1510b** may amplify the signal **1517b**. The modulator **1512b** may modulate the amplified signal **1517b** based on the chopping frequency f_{chop} . The ADC **1514b** may digitize the modulated signal **1517b**. The comb filter **1516b** may filter the digitized signal **1517b**.

(126) The CORDIC module **1522** may include any suitable type of processing circuitry that is configured to execute a Coordinate Rotation Digital Computer (CORDIC) algorithm or otherwise compute an arctangent function (e.g., such as by using a look-up table). According to the example of FIG. 15, the CORDIC module is configured to calculate a raw position signal based on the filtered signal **1517a** and the filtered signal **1517b**. The raw position signal may identify the orientation of the ring magnet **1420** relative to the sensor **1300**, and it may be indicative of angular displacement and/or rotational speed of the ring magnet **1420** (and/or the rotating shaft **1430**). In some implementations, the raw position signal may be calculated in accordance with Equation 10 below:

$$(127) \ S_{\text{raw}} = \arctan\left(\frac{\text{signal}_{1517a}}{\text{signal}_{1517b}}\right) \quad (\text{Eq. 10})$$

where S_{raw} is the raw position signal, signal_{1517a} is signal **1517a**, and signal_{1517b} is signal **1517b**.

(128) The error correction module **1524** may include any suitable type of processing circuitry for adjusting the gain and/or offset of the raw position signal that is produced by the CORDIC module **1522**. In operation, the error correction module **1524** may receive the raw position signal from the CORDIC module **1522** and generate an adjusted signal based on the received raw position signal. The adjusted signal may be generated by adjusting the gain and/or offset of the raw position signal. The gain and/or offset of the raw position signal may be adjusted, in a well-known fashion, based on a signal **1533** that is generated by a temperature sensor **1532**. Additionally or alternatively, the angle gain and/or offset of the raw position signal may be adjusted based on a signal **1570** that is generated by a trim module **1534**. The trim module **1534** may be a memory that is arranged to provide (to the error correction module) one or more coefficients for adjusting the gain and/or offset of the raw position signal. However, alternative implementations are possible in which the trim module **1534** includes another type of device (e.g., a humidity sensor, etc.) that is used for correcting the gain and/or offset of the raw position signal. Stated succinctly, the present disclosure is not limited to any specific method for adjusting the gain and/or offset of the raw position signal.

(129) The output module **1526** may include any suitable type of communications interface for outputting the adjusted signal that is produced by the error correction module **1524**. The output module may format the adjusted signal into a desired output signal format and provide the formatted signal to another device (e.g., an Engine Control Unit) that is coupled to the output module **1526**. The desired format may be PWM format, Single Edge Nibble Transmission (SENT) format, a Serial Peripheral Interface (SPI) format, a Local Interconnect Network (LIN) format, a

CAN (Controller Area Network) format, an Inter-Integrated Circuit (I^{sup}.2C) format to name a few non-limiting examples.

(130) FIG. 15 is provided in the context of the implementation of the sensor 1300 that is shown in FIG. 13A, in which the sensor 1300 is provided with four pairs of sensing elements. In the example of FIG. 15, the signal S.sub.raw is generated by taking the arctan of the quotient of signals 1517a and 1517b, where signals 1517a and 1517b are generated in accordance with equations 11 and 12 below:

$$\text{signal.sub.1517a} = (\text{signal.sub.1501a} - \text{signal.sub.1501d}) - (\text{signal.sub.1501b} + \text{signal.sub.1501c})$$

(11)

$$\text{signal.sub.1517b} = (\text{signal.sub.1503a} + \text{signal.sub.1503d}) - (\text{signal.sub.1503b} + \text{signal.sub.1503c})$$

(12)

where signal.sub.1501a is the signal output from the vertical Hall element 1304A, signal.sub.1501b is the signal output from the vertical Hall element 1304B, signal.sub.1501c is the signal output from the vertical Hall element 1304C, signal.sub.1501d is the signal output from the vertical Hall element 1304D, signal.sub.1503a is the signal output from the vertical Hall element 1302A, signal.sub.1503b is the signal output from the vertical Hall element 1302B, signal.sub.1503c is the signal output from the vertical Hall element 1302C, signal.sub.1503d is the signal output from the vertical Hall element 1302D. The signals 1517a and 1517b, which are generated in accordance with Equations 11 and 12, may be used to generate the signal Sraw, as discussed above with respect to Equation 10. As can be readily appreciated, equations 11-12 are provided for illustrative purposes only, and they do not reflect demodulation, amplification, filtering, and/or any other signal processing that might be performed.

(131) FIG. 15 is provided to illustrate one possible example of processing circuitry for the implementation of the sensor 1300, which is shown in FIG. 13A. In this regard, it will be understood that the present disclosure is not limited to any specific type of processing circuitry being used in the sensor 1300. For example, in some implementations, the number of modulators in each of the processing paths 1502a-b may be reduced from 4 to 2 by first subtracting the signals produced by each of the vertical Hall element pairs [1304A, 1304D], [1304B, 1304C], [1302A, 1302D], and [1302B, 1302C], and then demodulating the differences.

(132) FIG. 16 is a diagram of another implementation of the sensor 1300, according to aspects of the disclosure. As illustrated, in the example of FIG. 16, the sensor 1300 includes vertical Hall elements 1302B, 1302D, 1304B, and 1304D. The vertical Hall elements 1302B, 1302D, 1304B, and 1304D may be arranged in the manner discussed above with respect to FIG. 13A. FIG. 16 is provided to illustrate that the present disclosure is not limited to including any number of vertical Hall element pairs and that some of the vertical Hall element pairs shown in FIG. 13A can be omitted.

(133) FIG. 17 is a circuit diagram of a processing circuitry 1710 that can be used in the implementation of the sensor 1300 that is shown in FIG. 16. FIG. 17 is provided to illustrate one possible example of processing circuitry for the implementation of the sensor 1300, which is shown in FIG. 16. In this regard, it will be understood that the present disclosure is not limited to any specific arrangement of the processing circuitry of sensor 1300.

(134) The processing circuitry 1710 may include a processing path 1702a and a processing path 1702b. The processing path 1702a may be arranged to process signals that are generated by the vertical Hall elements 1304B and 1304D and the processing path 1702b may be arranged to process signals that are generated by the vertical Hall elements 1302B and 1302D.

(135) Processing path 1702a may include vertical Hall elements 1304B and 1304D, demodulators 1705b and 1705d, a subtraction element 1713a, an amplifier 1710a, a modulator 1712a, an analog to digital converter (ADC) 1714a, and a comb filter 1716a. Vertical Hall element 1304D may generate a signal 1701d, which is subsequently modulated by the modulator 1705d. Vertical Hall element 1304B may generate a signal 1701b, which is subsequently modulated by the modulator

1705b. The demodulation of each of signals **1701d** and **1701b** may be performed based on a chopping frequency **fchop**. After the signals **1701d** and **1701b** are demodulated, modulated signal **1701b** may be subtracted from modulated signal **1701d** to produce a signal **1717a**. The subtraction may be performed by using the subtraction element **1713a**. The amplifier **1710a** may amplify the signal **1717a**. The modulator **1712a** may modulate the amplified signal **1717a** based on the chopping frequency **fchop**. The ADC **1714a** may digitize the modulated signal **1717a**. The comb filter **1716a** may filter the digitized signal **1717a**.

(136) Processing path **1702b** may include vertical Hall elements **1302B** and **1302D**, demodulators **1707b** and **1707d**, a subtraction element **1713b**, an amplifier **1710b**, a modulator **1712b**, an analog-to-digital converter **1714b**, and a comb filter **1716b**. Vertical Hall element **1302D** may generate a signal **1703d**, which is subsequently modulated by the modulator **1707d**. Vertical Hall element **1302B** may generate a signal **1703b**, which is subsequently modulated by the modulator **1707b**. The demodulation of each of signals **1703b** and **1703d** may be performed based on the chopping frequency **fchop**. After the signals **1703b** and **1703d** are demodulated, modulated signal **1703b** may be subtracted from modulated signal **1703d** to produce a signal **1717b**. The subtraction may be performed by using subtraction element **1713b**. The amplifier **1710b** may amplify the signal **1717b**. The modulator **1712b** may modulate the signal **1717b** based on the chopping frequency **fchop**. The ADC **1714b** may digitize the modulated signal **1717b**. The comb filter **1716b** may filter the digitized signal **1717b**.

(137) The CORDIC module **1722** may include any suitable type of processing circuitry that is configured to execute a Coordinate Rotation Digital Computer (CORDIC) algorithm or otherwise compute an arctangent function (e.g., such as by using a look-up table). According to the example of FIG. 17, the CORDIC module is configured to calculate a raw position signal based on the filtered signal **1717a** and the filtered signal **1717b**. The raw position signal may identify the orientation of the ring magnet **1420** relative to the sensor **1300**, and it may be indicative of angular displacement and/or rotational speed of the ring magnet **1420** (and/or the rotating shaft **1430**). In some implementations, the raw position signal may be calculated in accordance with Equation 13 below:

$$(138) \quad S_{\text{raw}} = \arctan\left(\frac{\text{signal}_{1717a}}{\text{signal}_{1717b}}\right) \quad (\text{Eq. 13})$$

where $S_{\text{sub.raw}}$ is the raw position signal, signal.sub.1717a is signal **1717a**, and signal.sub.1717b is signal **1717b**.

(139) The error correction module **1724** may include any suitable type of processing circuitry for adjusting the gain and/or offset of the raw position signal that is produced by the CORDIC module **1722**. In operation, the error correction module **1724** may receive the raw position signal from the CORDIC module **1722** and generate an adjusted signal based on the received raw position signal. The adjusted signal may be generated by adjusting the gain and/or offset of the raw position signal. The gain and/or offset of the raw position signal may be adjusted, in a well-known fashion, based on a signal **1733** that is generated by a temperature sensor **1732**. Additionally or alternatively, the gain and/or offset of the raw position signal may be adjusted based on a signal **1770** that is generated by a trim module **1734**. The trim module **1734** may be a memory that is arranged to provide (to the error correction module) one or more coefficients for adjusting the gain and/or offset of the raw position signal. However, alternative implementations are possible in which the trim module **1734** includes another type of device (e.g., a humidity sensor, etc.) that is used for correcting the gain and/or offset of the raw position signal. Stated succinctly, the present disclosure is not limited to any specific method for adjusting the gain and/or offset of the raw position signal.

(140) The output module **1726** may include any suitable type of communications interface for outputting the adjusted signal that is produced by the error correction module **1724**. The output block may format the adjusted signal into a desired output signal format and provide the formatted signal to another device (e.g., an Engine Control Unit) that is coupled to the output module **1726**.

The desired format may be PWM format, Single Edge Nibble Transmission (SENT) format, a Serial Peripheral Interface (SPI) format, a Local Interconnect Network (LIN) format, a CAN (Controller Area Network) format, an Inter-Integrated Circuit (I.sup.2C) format to name a few non-limiting examples.

(141) FIG. 17 is provided in the context of the implementation of the sensor 1300 that is shown in FIG. 16, in which the sensor 1300 is provided with two pairs of sensing elements, instead of four. In the example of FIG. 17, the signal S.sub.raw is generated by taking the arctan of the quotient of signals 1717a and 1717b, where signals 1717a and 1717b are generated in accordance with equations 14 and 15 below:

$$\text{signal.sub.1717} = \text{signal.sub.1701d} - \text{signal.sub.1701b} \quad (14)$$

$\text{signal.sub.1717b} = \text{signal.sub.1703d} - \text{signal.sub.1703b}$ (15) where signal.sub.1701d is the signal output from the vertical Hall element 1304D, signal.sub.1701b is the signal output from the vertical Hall element 1304B, signal.sub.1703d is the signal output from the vertical Hall element 1302D, and signal.sub.1703b is the signal output from the vertical Hall element 1302B. The signals 1717a and 1717b, which are generated in accordance with Equations 14 and 15, may be used to generate the signal Sraw, as discussed above with respect to Equation 13. As can be readily appreciated, equations 14-15 are provided for illustrative purposes only, and they do not reflect demodulation, amplification, filtering, and/or any other signal processing that might be performed.

(142) The processes described herein may be implemented in hardware, software, or a combination of the two. The processes described herein may be implemented in computer programs executed on programmable computers/machines that each includes a processor, a non-transitory machine-readable medium or another article of manufacture that is readable by the processor (including volatile and non-volatile memory and/or storage elements), at least one input device, and one or more output devices. Program code may be applied to data entered using an input device to perform any of the processes described herein and to generate output information.

(143) The system may be implemented, at least in part, via a computer program product, (e.g., in a non-transitory machine-readable storage medium such as, for example, a non-transitory computer-readable medium), for execution by, or to control the operation of, data processing apparatus (e.g., a programmable processor, a computer, or multiple computers). Each such program may be implemented in a high-level procedural or object-oriented programming language to work with the rest of the computer-based system. However, the programs may be implemented in assembly, machine language, or Hardware Description Language. The language may be a compiled or an interpreted language, and it may be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or another unit suitable for use in a computing environment. A computer program may be deployed to be executed on one computer or multiple computers at one site or distributed across multiple sites and interconnected by a communication network. A computer program may be stored on a non-transitory machine-readable medium that is readable by a general or special purpose programmable computer for configuring and operating the computer when the non-transitory machine-readable medium is read by the computer to perform the processes described herein. For example, the processes described herein may also be implemented as a non-transitory machine-readable storage medium, configured with a computer program, where upon execution, instructions in the computer program cause the computer to operate in accordance with the processes. A non-transitory machine-readable medium may include but is not limited to a hard drive, compact disc, flash memory, non-volatile memory, volatile memory, magnetic diskette and so forth but does not include a transitory signal per se.

(144) Having described preferred embodiments, which serve to illustrate various concepts, structures and techniques, which are the subject of this patent, it will now become apparent that other embodiments incorporating these concepts, structures and techniques may be used. Accordingly, it is submitted that the scope of the patent should not be limited to the described embodiments but rather should be limited only by the spirit and scope of the following claims.

Claims

1. A sensor, comprising: a substrate having a reference crystal orientation; and a plurality of vertical Hall element pairs that are formed on the substrate, each vertical Hall element pair including: (i) a respective first vertical Hall element that is oriented at a respective first angle relative to the reference crystal orientation of the substrate and (ii) a respective second vertical Hall element that is oriented at a respective second angle relative to the reference crystal orientation of the substrate, wherein the substrate has a rectangular shape, and each of the vertical Hall element pairs is disposed in a different respective corner of the substrate, wherein the respective first vertical Hall element in at least two of the vertical Hall element pairs is oriented at substantially +45 degrees relative to the reference crystal orientation of the substrate, and the respective second vertical Hall element in at least two of the vertical Hall element pairs is oriented at substantially -45 degrees relative to the reference crystal orientation of the substrate.
2. The sensor of claim 1, wherein; the vertical Hall elements in each of the vertical Hall element pairs have axes of maximum sensitivity that are substantially perpendicular to each other, one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially parallel with a given diagonal of the substrate, and the other one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially perpendicular to the given diagonal of the substrate.
3. The sensor of claim 1, wherein: the respective first vertical Hall element in each of the vertical Hall element pairs is configured to face towards the vertical Hall element pair's respective corner, and the respective second vertical Hall element in each of the vertical Hall element pairs is configured to face away from the vertical Hall element pair's respective corner.
4. The sensor of claim 1, wherein: one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially parallel with a given diagonal of the substrate, and the other one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially perpendicular to the given diagonal of the substrate.
5. The sensor of claim 1, wherein the plurality of vertical Hall element pairs includes at least two vertical Hall element pairs.
6. The sensor of claim 1, further comprising a processing circuitry configured to: receive a plurality of signals, each of the signals being generated by a different respective vertical Hall element in any of the plurality of vertical Hall element pairs, each of the signals being generated in response to a magnetic field that is associated with a rotating target; and generate an output signal that is indicative of at least one of speed or position of the rotating target.
7. The sensor of claim 1, wherein the substrate includes a silicon die.
8. A sensor, comprising: a substrate having a reference crystal orientation; and a plurality of vertical Hall element pairs that are formed on the substrate, each vertical Hall element pair including: (i) a respective first vertical Hall element that is oriented at a respective first angle relative to the reference crystal orientation of substrate and (ii) a respective second vertical Hall element that is oriented at a respective second angle relative to the reference crystal orientation of the substrate, wherein the substrate has a rectangular shape, and each of the vertical Hall element pairs is disposed in a different respective corner of the substrate, wherein the plurality of vertical Hall elements includes a first vertical Hall element pair, a second vertical Hall element pair, a third vertical Hall element pair, and a fourth vertical Hall element pair, the sensor further comprising processing circuitry configured to generate a differential signal having a first component and a second component, the first component and the second component being defined as follows:
firstComponent=S.sub.1,1-S.sub.4,1+S.sub.2,1-S.sub.3,1
secondComponent=S.sub.1,2-S.sub.4,2+S.sub.2,2-S.sub.3,2 where S.sub.1,1 is a signal that is

output by the respective first vertical Hall element of the first vertical Hall element pair, S.sub.4,1 is a signal that is output by the respective first vertical Hall element of the fourth vertical Hall element pair, S.sub.2,1 is a signal that is output by the respective first vertical Hall element of the second vertical Hall element pair, S.sub.3,1 is a signal that is output by the respective first vertical Hall element of the third vertical Hall element pair, S.sub.1,2 is a signal that is output by the respective second vertical Hall element of the first vertical Hall element pair, S.sub.4,2 is a signal that is output by the respective second vertical Hall element of the fourth vertical Hall element pair, S.sub.2,2 is a signal that is output by the respective second vertical Hall element of the second vertical Hall element pair, and S.sub.3,2 is a signal that is output by the respective second vertical Hall element of the third vertical Hall element pair.

9. The sensor of claim 8, wherein each of the signals S.sub.1,1, S.sub.4,1, S.sub.2,1, S.sub.3,1, S.sub.1,2, S.sub.4,2, S.sub.2,2, S.sub.3,2 is generated in response to a magnetic field associated with a rotating target, and the processing circuitry is configured to generate an output signal based on the differential signal, the output signal being indicative of at least one of speed or position of the rotating target.

10. A sensor, comprising: a substrate having a rectangular shape; a plurality of vertical Hall element pairs that are formed on the substrate, each of the vertical Hall element pairs being formed in a different corner of the substrate, each of the vertical Hall element pairs including: (i) a respective first vertical Hall element that is configured to face the vertical Hall element pair's respective corner, and (ii) a respective second vertical Hall element that is configured to face away from the vertical Hall element pair's respective corner, wherein the substrate has a reference crystal orientation, the respective first vertical Hall element in at least two of the vertical Hall element pairs is oriented at substantially +45 degrees relative to the reference crystal orientation of the substrate, and the respective second vertical Hall element in at least two of the vertical Hall element pairs is oriented at substantially -45 degrees relative to the reference crystal orientation of the substrate.

11. The sensor of claim 10, wherein: the vertical Hall elements in each of the vertical Hall element pairs have axes of maximum sensitivity that are substantially perpendicular to each other, one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially parallel with a given diagonal of the substrate, and the other one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially perpendicular to the given diagonal of the substrate.

12. The sensor of claim 10, wherein: the substrate has a reference crystal orientation, the respective first vertical Hall element in each of the vertical Hall element pairs is oriented at a respective first angle relative to the reference crystal orientation of the substrate, and the respective second vertical Hall element in each of the vertical Hall element pairs is oriented at a respective second angle relative to the reference crystal orientation of the substrate.

13. The sensor of claim 10, wherein: one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially parallel with a given diagonal of the substrate, and the other one of the vertical Hall elements in each of the vertical Hall element pairs has an axis of maximum sensitivity that is substantially perpendicular to the given diagonal of the substrate.

14. The sensor of claim 10, wherein the plurality of vertical Hall element pairs includes at least two vertical Hall element pairs.

15. A sensor, comprising: a substrate having a rectangular shape: a plurality of vertical Hall element pairs that are formed on the substrate, each of the vertical Hall element pairs being formed in a different corner of the substrate, each of the vertical Hall element pairs including: (i) a respective first vertical Hall element that is configured to face the vertical Hall element pair's respective corner, and (ii) a respective second vertical Hall element that is configured to face away from the vertical Hall element pair's respective corner, wherein the plurality of vertical Hall elements

includes a first vertical Hall element pair, a second vertical Hall element pair, a third vertical Hall element pair, and a fourth vertical Hall element pair, the sensor further comprising processing circuitry configured to generate a differential signal having a first component and a second component, the first component and the second component being defined as follows:

$\text{firstComponent} = S_{\text{sub.1,1}} - S_{\text{sub.4,1}} + S_{\text{sub.2,1}} - S_{\text{sub.3,1}}$

$\text{secondComponent} = S_{\text{sub.1,2}} - S_{\text{sub.4,2}} + S_{\text{sub.2,2}} - S_{\text{sub.3,2}}$ where $S_{\text{sub.1,1}}$ is a signal that is output by the respective first vertical Hall element of the first vertical Hall element pair, $S_{\text{sub.4,1}}$ is a signal that is output by the respective first vertical Hall element of the fourth vertical Hall element pair, $S_{\text{sub.2,1}}$ is a signal that is output by the respective first vertical Hall element of the second vertical Hall element pair, $S_{\text{sub.3,1}}$ is a signal that is output by the respective first vertical Hall element of the third vertical Hall element pair, $S_{\text{sub.1,2}}$ is a signal that is output by the respective second vertical Hall element of the first vertical Hall element pair, $S_{\text{sub.4,2}}$ is a signal that is output by the respective second vertical Hall element of the fourth vertical Hall element pair, $S_{\text{sub.2,2}}$ is a signal that is output by the respective second vertical Hall element of the second vertical Hall element pair, and $S_{\text{sub.3,2}}$ is a signal that is output by the respective second vertical Hall element of the third vertical Hall element pair.

16. The sensor of claim 15, wherein each of the signals $S_{\text{sub.1,1}}$, $S_{\text{sub.4,1}}$, $S_{\text{sub.2,1}}$, $S_{\text{sub.3,1}}$, $S_{\text{sub.1,2}}$, $S_{\text{sub.4,2}}$, $S_{\text{sub.2,2}}$, $S_{\text{sub.3,2}}$ is generated in response to a magnetic field associated with a rotating target, and the processing circuitry is configured to generate an output signal based on the differential signal, the output signal being indicative of at least one of speed or position of the rotating target.

17. The sensor of claim 10, further comprising a processing circuitry configured to: receive a plurality of signals, each of the signals being generated by a different respective vertical Hall element in any of the plurality of vertical Hall element pairs, each of the signals being generated in response to a magnetic field that is associated with a rotating target; and generate an output signal that is indicative of at least one of speed or position of the rotating target.

18. A system, comprising: a ring magnet having first surface, a second surface, and a bore extending from the first surface to the second surface, the bore having a central longitudinal axis; a substrate having a reference crystal orientation; a plurality of vertical Hall element pairs that are formed on the substrate, each vertical Hall element pair including: (i) a respective first vertical Hall element that is oriented at a respective first angle relative to the reference crystal orientation of the substrate and (ii) a respective second vertical Hall element that is oriented at a respective second angle relative to the reference crystal orientation of the substrate, wherein the respective first vertical Hall element in at least two of the vertical Hall element pairs is often at substantially +45 degrees relative to the reference crystal orientation of the substrate, and the respective second vertical Hall element in at least two of the vertical Hall element pairs is oriented at substantially -45 degrees relative to the reference crystal orientation of the substrate.

19. The system of claim 18, wherein the substrate is inserted into the bore of the ring magnet.

20. The system of claim 18, wherein the vertical Hall elements in each of the vertical Hall element pairs have axes of maximum sensitivity that are substantially perpendicular to each other.

21. The system of claim 18, wherein: the substrate has a rectangular shape; each of the vertical Hall element pairs is disposed in a different respective corner of the substrate, the respective first vertical Hall element in each of the vertical Hall element pairs is configured to face towards the vertical Hall element pair's respective corner, and the respective second vertical Hall element in each of the vertical Hall element pairs is configured to face away from the vertical Hall element pair's respective corner.

22. The system of claim 18, further comprising a processing circuitry configured to: receive a plurality of signals, each of the signals being generated by a different respective vertical Hall element in any of the plurality of vertical Hall element pairs, each of the signals being generated in response to a magnetic field that is associated with a rotating target; and generate an output signal

that is indicative of at least one of speed or position of the rotating target.

23. A system, comprising: a ring magnet having first surface, a second surface, and a bore extending from the first surface to the second surface, the bore having a central longitudinal axis, a substrate having a reference crystal orientation; a plurality of vertical Hall element pairs that are formed on the substrate, each vertical Hall element pair including: (i) a respective first vertical Hall element that is oriented at a respective first angle relative to the reference crystal orientation of the substrate and (ii) a respective second vertical Hall element that is oriented at a respective second angle relative to the reference crystal orientation of the substrate, wherein the plurality of vertical Hall elements includes a first vertical Hall element pair, a second vertical Hall element pair, a third vertical Hall element pair, and a fourth vertical Hall element pair, the system further comprising processing circuitry configured to generate a differential signal having a first component and a second component, the first component and the second component being defined as follows:

$$\text{firstComponent} = S_{\text{sub.1,1}} - S_{\text{sub.4,1}} + S_{\text{sub.2,1}} - S_{\text{sub.3,1}}$$

$$\text{secondComponent} = S_{\text{sub.1,2}} - S_{\text{sub.4,2}} + S_{\text{sub.2,2}} - S_{\text{sub.3,2}}$$
 where $S_{\text{sub.1,1}}$ is a signal that is output by the respective first vertical Hall element of the first vertical Hall element pair, $S_{\text{sub.4,1}}$ is a signal that is output by the respective first vertical Hall element of the fourth vertical Hall element pair, $S_{\text{sub.2,1}}$ is a signal that is output by the respective first vertical Hall element of the second vertical Hall element pair, $S_{\text{sub.3,1}}$ is a signal that is output by the respective first vertical Hall element of the third vertical Hall element pair, $S_{\text{sub.1,2}}$ is a signal that is output by the respective second vertical Hall element of the first vertical Hall element pair, $S_{\text{sub.4,2}}$ is a signal that is output by the respective second vertical Hall element of the fourth vertical Hall element pair, $S_{\text{sub.2,2}}$ is a signal that is output by the respective second vertical Hall element of the second vertical Hall element pair, and $S_{\text{sub.3,2}}$ is a signal that is output by the respective second vertical Hall element of the third vertical Hall element pair.

24. The system of claim 23, wherein each of the signals $S_{\text{sub.1,1}}$, $S_{\text{sub.4,1}}$, $S_{\text{sub.2,1}}$, $S_{\text{sub.3,1}}$, $S_{\text{sub.1,2}}$, $S_{\text{sub.4,2}}$, $S_{\text{sub.2,2}}$, $S_{\text{sub.3,2}}$ is generated in response to a magnetic field associated with a rotating target, and the processing circuitry is configured to generate an output signal based on the differential signal, the output signal being indicative of at least one of speed or position of the rotating target.
